

Abstract Algebra

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PREFACE

These reading notes are a combination of material from the courses Math 210A by Prof. Richard Taylor at Stanford University, Dummit and Foote's *"Abstract Algebra"* [1] and Aluffi's *"Algebra Chapter 0"* [2].

There may be major and minor errors throughout these notes. If you find any, please let me know by sending me an email at jubayer@stanford.edu. Oftentimes, I wrote these notes *after* I had already handwritten them elsewhere and I was careless / inefficient in rewriting the proof. I also glossed over various parts of the original texts that the reading notes were trying to follow despite them being pretty important. As such, I take sole responsibility for the errors.

1. GROUPS

1.1. DEFINITIONS AND IMMEDIATE PROPERTIES

Definition 1.1 (Groups). A **group** is the following data: G is a set and $\bullet$ is a binary operation on G , such that:

1. the operation is associative: $(x \bullet y) \bullet z = x \bullet (y \bullet z)$ for all $x, y, z \in G$
2. the operation has an identity: there exists an element $1_G \in G$ such that $x \bullet 1_G = 1_G \bullet x = x$ for all $x \in G$
3. the operation gives each element an inverse: for all $x \in G$, there exists an inverse $x^{-1} \in G$ such that $x \bullet x^{-1} = x^{-1} \bullet x = 1_G$

In particular, the operation $\bullet$ need not be commutative. However, if it is commutative (i.e. $x \bullet y = y \bullet x$ for all $x, y \in G$), then we say that the group is **abelian** or commutative.

Observe that a group is always non-empty since it must, at the very least, have the identity, $1_G \in G$. We now look at some immediate examples of groups:

Examples 1.2. The following are all groups:

1. The set of integers ($\mathbb{Z}$), the set of rationals ($\mathbb{Q}$), the set of reals ($\mathbb{R}$), and the set of complex numbers ($\mathbb{C}$) are all groups under addition, and they are all abelian. However, they are not all groups under multiplication. For example, $\mathbb{Z}$ is not a group under multiplication since $x > 1$ and $x < -1$ has no inverse. Furthermore, $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are not groups under multiplication since 0 has no inverse.
2. The sets $\mathbb{R} - \{0\}$, $\mathbb{Q} - \{0\}$, and $\mathbb{C} - \{0\}$ are groups under multiplication (but not under addition since there is no additive identity).
3. All vector spaces V are groups under addition. In particular, they are abelian groups.
4. The set $\mathbb{Z}/n\mathbb{Z}$ where $n \in \mathbb{Z}_{>0}$ is an abelian group under addition. The identity in the group is $\bar{0}$ and the additive inverse of $\bar{x}$ is $-\bar{x}$ for any $\bar{x} \in \mathbb{Z}/n\mathbb{Z}$.

Note that $\mathbb{Z}/n\mathbb{Z}$ is not guaranteed to be a group under multiplication since, a priori, all elements are not guaranteed to have a multiplicative inverse. First, we define the subset of $\mathbb{Z}/n\mathbb{Z}$ which admits such inverses to be a group:

Example 1.3. The set $(\mathbb{Z}/n\mathbb{Z})^\times \subseteq \mathbb{Z}/n\mathbb{Z}$ is the set of all equivalence classes $\bar{x}$ such that there exists a multiplicative inverse. This is a group under multiplication where the identity is $\bar{1}$.

It should be clear, from the notation itself, that $\mathbb{Z}/n\mathbb{Z}$ is a group under addition and $(\mathbb{Z}/n\mathbb{Z})^\times$ is a group under multiplication.

Definition 1.4 (Direct Product of Groups). Suppose $(X, \bullet)$ and $(Y, \square)$ are groups. Then, the direct product of these groups is the set $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$ under the componentwise operation:

$$(x_1, y_1)(x_2, y_2) = (x_1 \bullet x_2, y_1 \square y_2).$$

We now look at some immediate properties that all groups satisfy.

Notation 1.5. We adopt the shorthand notation where we suppress the group operation $x_1 \bullet x_2$ and write it as $x_1 x_2$. We write multiplicatively even though the group operation need not be.

Theorem 1.6. Suppose G is a group. Then,

1. the identity 1_G of G is unique

2. the inverse x^{-1} of each $x \in G$ is unique
3. $(x^{-1})^{-1} = x$ for all $x \in G$
4. $(xy)^{-1} = y^{-1}x^{-1}$ for all $x, y \in G$
5. If $xy = xz$ for some $x, y, z \in G$, then $y = z$. If $yx = zx$, then $y = z$. In particular, there is a unique solution for $ax = b$ in x and a unique solution for $ya = b$ in y .

Finally, we define the *order* of an element.

Definition 1.7 (Order of an Element in a Group). *Let G be a group and let $x \in G$. Then, the order of x is denoted by $|x|$ and is defined as the smallest positive integer n such that $x^n = 1_G$. If there is no such n , then we say that the order is infinite.*

Examples 1.8. 1. The element $1_G \in G$ has order 1. In fact, it is the element with this property, i.e. an element in any group has order 1 if and only if it is the identity.

2. In $\mathbb{R} - \{0\}$, or $\mathbb{Q} - \{0\}$, the element -1 has order 2 under the operation of multiplication. All other elements in these groups have infinite order.

We briefly also look at the definitions of monoids.

Definition 1.9 (Monoids). *A monoid is a triple $(M, \cdot, 1)$ where M is a non-empty set, $\cdot$ is a binary operation that is associative (i.e. $x \cdot (y \cdot z) = (x \cdot y) \cdot z$ for all $x, y, z \in M$) and $1 \in M$ such that $1 \cdot m = m = m \cdot 1$ for all $m \in M$.*

In other words, a group is a monoid $(G, \cdot, 1)$ where every element in G has an inverse. We also often encounter *monads*, which has the same definition as a monoid except the binary operation need not be associative. On the other hand, a semigroup $(M, \cdot)$ is a non-empty set with an associative binary operation that need not have the identity 1.

1.2. SYMMETRIC GROUPS

The symmetric group is an extraordinarily important and ubiquitous construction.

Definition 1.10 (Symmetric Groups). *Let M be a set and let S_M be the set of all bijections from M to itself i.e. the set of all permutations of M . Then, S_M is a group under composition i.e. if $\sigma_s, \sigma_t \in S_M$, then $\sigma_s \sigma_t = \sigma_s \circ \sigma_t$. The identity of the group is 1 which sends every element in M to itself.*

In particular, when the set is $M = \{1, 2, \dots, n\}$ of size n , the symmetric group is denoted by S_n and called the symmetric group of degree n .

Sometimes, the symmetric group of the set M is denoted by $\text{Sym}(M)$. In these notes, however, we simply denote it by S_M .

Proposition 1.11. *The order of the symmetric group S_n is $n!$, i.e. $|S_n| = n!$.*

Proof. Each permutation in a symmetric group is an injective function from $\{1, \dots, n\}$ to itself. There are $n!$ such injective functions. $\square$

One can denote the elements of a symmetric group using cycles. A cycle is a string such as $(a_1, a_2, \dots, a_m)$ which represents a permutation; this cycle represents that the permutation takes a_1 to a_2 , a_2 to a_3 and so on until a_m to a_1 . The cycle $(3, 4, 1)$ takes 3 to 4, 4 to 1, and 1 to 3. With this notation, we can decompose any $\sigma \in S_n$ as a composition of cycles of the form $(a_1, a_2, \dots, a_m)(a_p, a_{p+1}, \dots, a_{p+k})$. The length of a cycle is the number of integers in it. In particular, we have the following method for calculating the order of any element in a symmetric group:

Proposition 1.12. *The order of a permutation in a symmetric group is the l.c.m. of the lengths of the cycles in its cycle decomposition.*

Proposition 1.13. *S_n is a non-abelian group for all $n \geq 3$.*

Proof. Consider the set $\{1, 2, 3, \dots, n\}$. Then, consider the two permutations σ_s defined by $1 \rightarrow 3$ and the permutation σ_t defined by $1 \rightarrow 2$. Then, $\sigma_s \sigma_t \neq \sigma_t \sigma_s$. Since these permutations exist in all symmetric groups S_n for $n \geq 3$, we are done. $\square$

1.3. DIHEDRAL GROUPS

Dihedral groups are an extremely important class of groups that define symmetries of polygons.

Definition 1.14 (Dihedral groups). *For a positive integer $n \geq 3$, the dihedral group D_{2n} is the set of all symmetries of the regular n -gon. A symmetry is defined as any rigid motion on the object such that the objects look exactly the same. More precisely:*

- *A symmetry of an n -gon is the following: label all n vertices of the polygon first. Then, a symmetry is a permutation σ of the set $\{1, 2, \dots, n\}$ corresponding to the rigid motion that leaves the polygon unchanged. For e.g., consider the 90 degrees clockwise rotation of a square (i.e. 4-gon) whose vertices are labelled sequentially. Then, the rotation can be described as the permutation σ that sends i to $i + 1$, for all $i \in \{1, \dots, n - 1 = 3\}$, and $n = 4$ to 1.*

This set D_{2n} is a group under the operation of composition. In other words, if $s, t \in D_{2n}$ (i.e. s is some symmetry and t is another symmetry), then st is the symmetry obtained by first applying t and then applying s . This can also be defined by looking at the permutations each such symmetry corresponds to: if s corresponds to the permutation σ_s and t corresponds to the permutation σ_t , then st corresponds to the permutation $\sigma_s \circ \sigma_t$.

In D_{2n} , the identity is the identity symmetry which leaves all vertices fixed/unchanged and is denoted by 1. The inverse of $s \in D_{2n}$ is the symmetry that reverses all the rigid motions of s i.e. s^{-1} corresponds to the permutation σ^{-1} if s corresponds to σ .

Proposition 1.15. *For any dihedral group D_{2n} , the size of the set is $|D_{2n}| = 2n$. We say the dihedral group D_{2n} is of order $2n$.*

Proof. Consider some sequential numbering of the vertices of the n -gon i.e. 1 is next to 2 which is next to 3 and so on. Now, we can send the vertex number 1 to n possible locations/vertices. Suppose we send 1 to i . Then, we must send 2 to either $i + 1$ or $i - 1$ because symmetries must be rigid motions. That means we have $n \times 2 = 2n$ possible permutations that define symmetries. $\square$

Definition 1.16 (Rotations and Reflections in D_{2n}). *Define $r \in D_{2n}$ to be a clockwise rotation about the origin through $\frac{2\pi}{n}$ radians. Define $s \in D_{2n}$ to be reflection about the line going through the vertex 1 and the origin.*

Proposition 1.17. *The following are satisfied in a dihedral group:*

1. $1, r, r^2, \dots, r^{n-1}$ are distinct and $r^n = 1$. In particular, $|r| = n$
2. $|s| = 2$
3. $s \neq r^i$ for any i
4. $sr^i \neq sr^j$ for $i \neq j$ and $0 \leq i, j \leq n - 1$

As such, we can write out the full group as:

$$D_{2n} = \{1, r, r^2, \dots, r^{n-1}, s, sr, sr^2, \dots, sr^{n-1}\}$$

and any $x \in D_{2n}$ can be written as $x = s^i r^j$ for $i \in \{0, 1\}$ and $0 \leq j \leq n-1$. Furthermore,

1. $rs = sr^{-1}$. As such, **the dihedral group is not an abelian group**.
2. $r^i s = sr^{-i}$ for all $0 \leq i \leq n$.

1.4. HOMOMORPHISMS AND ISOMORPHISMS

A recurring, central question we ask in abstract algebra is whether or not two mathematical objects of interest are, in some sense, the same. To be able to ask this question, we define homomorphisms and isomorphisms that help us understand these questions for groups with respect to their operations.

Definition 1.18 (Homomorphisms and Isomorphisms). Let $(G, \bullet)$ and $(H, \square)$ be two groups. Then, $\varphi : G \rightarrow H$ is a **homomorphism** if

$$\varphi(x \bullet y) = \varphi(x) \square \varphi(y)$$

for all $x, y \in G$. In other words, a homomorphism is a mapping that respects the structure of the group G and the group H . In particular, if φ is a bijection, then we call it an **isomorphism**, and we write $G \cong H$.

We will switch to the more convenient notation and simply write $\varphi(xy) = \varphi(x)\varphi(y)$ by suppressing the group operation.

Recall that S_M is the symmetric group over the set M . We now show that symmetric groups over sets of equal size are isomorphic.

Theorem 1.19. Let M and M' be two sets. Then, $S_M \cong S_{M'}$ if and only if $|M| = |M'|$.

Proof Sketch: The forward direction is simple. First, since both the sets have the same cardinality, define some correspondence of the elements between them via θ i.e. $\theta(m) = m'$ for some $m \in M, m' \in M'$ where θ is bijective. This map, θ , is our way of associating each element in M with an element in M' which we can do since they have the same cardinality. Now, suppose $\sigma(x) = y$ for $x, y \in M$, i.e. σ is an element in S_M . Then, we will map this to a symmetry in $S_{M'}$ by looking at how the corresponding elements in M' get permuted. Define $\varphi : S_M \rightarrow S_{M'}$ by $\varphi(\sigma)(\theta(x)) = \theta(y)$. In other words, φ takes σ to a permutation in $S_{M'}$ that moves the elements of M' the same way σ moves the elements of M . One can easily show that this is a homomorphism. Furthermore, this is invertible since θ itself is invertible, i.e. given any $\mu \in S_{M'}$, we can find a corresponding $\sigma \in S_M$ using the map θ . The converse direction is easy to prove for the finite group case (we will prove the infinite case later). Since the two groups are isomorphic, it must be that $|S_M| = |S_{M'}|$. now, suppose $|M| = p$ and $|M'| = q$. Then, $|S_M| = p!$ and $|S_{M'}| = q!$. So, $p = q$ and we are done.

Examples 1.20. The following are examples of isomorphisms:

1. $G \cong G$ for any group G via, at least, the identity isomorphism.
2. Any non-abelian group of order 6 is isomorphic to S_3 . In particular, $D_6 \cong S_3$. We shall prove this later when we study classification theorems.
3. $\varphi : \mathbb{R} \rightarrow \mathbb{R}_{>0}$ by $\varphi(x) = \exp(x)$ is an isomorphism from $(\mathbb{R}, +)$ to $(\mathbb{R}_{>0}, \times)$.

Proposition 1.21. If $\varphi : G \rightarrow H$ is an isomorphism, then

1. $|G| = |H|$

2. G is abelian if and only if H is abelian
3. $|x| = |\varphi(x)|$ for all $x \in G$.

Example 1.22. Note that if $\varphi : G \rightarrow H$ is an isomorphism, then $|G| = |H|$, but the converse is not true. For example, S_3 is not isomorphic to $\mathbb{Z}/6\mathbb{Z}$, despite their cardinality being the same, because the latter is abelian while the former is not.

1.5. GROUP ACTIONS

So far, we have seen homomorphisms/isomorphisms which maps a group G to a group H . Now, we will look at a particular mapping from $G \times A$ to A where G is a group and A is an arbitrary set.

Definition 1.23 (Group Action). A **group action** of a group G on a set A is a mapping $\varphi : G \times A \rightarrow A$ such that:

1. $\varphi(g_1, \varphi(g_2, a)) = \varphi(g_1 g_2, a)$ for all $g_1, g_2 \in G, a \in A$
2. $\varphi(1, a) = a$ for all $a \in A$

We will usually use notation that suppresses the fact that a group action is a mapping and, instead, write ga to mean $\varphi(g, a)$. As such, the requirements can be written as $g_1(g_2 a) = (g_1 g_2)a$ and $1a = a$.

Observation. A group action can be thought of as a permutation of the set. Since each group element maps elements $a \in A$ to $a' \in A$, we can think of each group element as a permutation in the symmetric group, S_A . Concretely, $g \cdot a$ is $g \in G$ is acting on $a \in A$. Now, for each $g \in G$, we can define $\sigma_g : A \rightarrow A$ by $\sigma_g(a) = g \cdot a$. Then,

1. σ_g is a permutation of A for each g , and
2. $\psi : G \rightarrow S_A$ via $\psi(g) = \sigma_g$ is a homomorphism. In particular, ψ is called the **permutation representation** associated to the given group action. In other words, for each group action, there exists a permutation representation corresponding to the group action.

To see the first, one can show that σ_g has an inverse which is $\sigma_{g^{-1}}$; this follows from the first defining property of a group action. To see the second, we can show that $\psi(g_1 g_2)(a) = (\psi(g_1) \cdot \psi(g_2))(a)$. This means, a group action of G on a set A is to be understood as a way of turning every element of G into a permutation on the set A .

Observation. We saw that, given a group action, there is an associated permutation representation i.e. a homomorphism $G \rightarrow S_A$. We can reverse this process. Given a homomorphism $\psi : G \rightarrow S_A$, we can construct a group action $G \times A \rightarrow A$ by setting $g \cdot a = \psi(g)(a)$.

Examples 1.24. We now look at a number of examples of group actions.

1. The trivial action is defined by setting $g \cdot a = a$ for all $a \in A$ and all $g \in G$ i.e. all elements of G correspond to the identity permutation in S_A . The associated permutation representation sends every element in G to the identity $1 \in S_A$.
2. On the other hand, if each element of G corresponds to a unique permutation in S_A , we say that the group action is **faithful**. The associated permutation representation is injective.
3. Let our group be a field F and let V be a vector space defined over F . Then, F acts on V via $\alpha \vec{v} = \alpha(v_1, \dots, v_n) = (\alpha v_1, \dots, \alpha v_n) \in V$.
4. Let A be a nonempty set. Then, the symmetric group of A , S_A , acts on A via $\sigma \cdot a = \sigma(a)$ for any $a \in A, \sigma \in S_A$. The associated permutation representation, $\text{id} : S_A \rightarrow S_A$, is identity.
5. After fixing some labeling of the vertices of an n -gon, each element of D_{2n} corresponds to a permutation σ in S_n defined by the way the element permutes the corresponding vertices. This gives us a group action $D_{2n} \times \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ i.e. $\sigma \cdot i = j$ where j is the permutation of i under σ .

1.6. SUBGROUPS

Definition 1.25 (Subgroups). Let G be a group and let H be a subset of G . Then, H is called a subgroup if

1. H is non-empty, i.e. $H \neq \emptyset$
2. H is closed under the group operation and inverses, i.e. $xy, x^{-1} \in H$ for all $x, y \in H$

In symbols, we write $H \leq G$.

Examples 1.26. We look at some examples and non-examples of subgroups.

1. Any group G has at least two subgroups: $H = G$ and $H = \{1_G\}$.
2. $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R}$.
3. Note that the subgroup must be defined with respect to the same operation as that of the group. For example, $(\mathbb{Q} - \{0\}, \times)$ is not a subgroup of $(\mathbb{R}, +)$ even though both are groups.
4. D_6 is not a subgroup of D_8 since D_6 is not a subset of D_8 - this is because the symmetries in D_6 are not the same as the symmetries in D_8 (e.g., the rotation is by 60 degrees in D_6 whereas it is by 45 degrees in D_8).

Alternatively, one can verify whether H is a subgroup of G via the following criterion:

Proposition 1.27 (Subgroup Criterion). A subset $H \subseteq G$ of a group G is a subgroup if and only if

1. $H \neq \emptyset$, and
2. $xy^{-1} \in H$ for all $x, y \in H$.

If H is finite, an equivalent and sufficient criterion is to simply check that

1. $H \neq \emptyset$
2. H is closed under group operation.

We will now look at some special classes of subgroups.

1.6.1 Centralizers, Normalizers, Centre

Given a subset A of a group G , we can construct some important classes of subgroups of G defined by how they interact with that specific set A . This leads us to the definitions of centralizers and normalizers. The centre of a group is another important subgroup that is defined by considering the subset $A = G$.

Definition 1.28 (Centralizers). Let G be a group and let A be a subset of G that is non-empty. Then, the centralizer of A in G is defined as

$$C_G(A) := \{g \in G \mid gag^{-1} = a, \forall a \in A\}.$$

In other words, the centralizer of A in G are **all the elements in G that elements of A commute with**.

Theorem 1.29. The centralizer, $C_G(A)$, is a subgroup, i.e. $C_G(A) \leq G$.

The proof is straightforward using the subgroup criterion.

Observation. Note that if G is an abelian group, then for any non-empty subset A of G , $C_G(A) = G$.

The centralizer is a particularly strict criterion in the sense that $a \in A$ must commute with all the elements in $C_G(A)$ i.e. $ag = ga$ for all $g \in C_G(A)$. Next, we look at the normalizer which relaxes the condition quite a bit.

Definition 1.30 (Normalizer). *let A be a non-empty subset of the group G . Then, the normalizer of A in G is the set*

$$N_G(A) := \{g \in G \mid gAg^{-1} = A\}$$

where $gAg^{-1} = \{gag^{-1} \mid a \in A\}$. In other words, the normalizer of A in G are all the elements in G that conjugate elements of A (i.e. gag^{-1}) and return some element in A .

So far, our constructions were with respect to some non-empty subset A of a group G . Next, we look at the centre of a group, which can be seen as the centralizer of the subset G of G .

Theorem 1.31. *The normalizer of a set A in group G is a subgroup, i.e. $N_G(A) \leq G$.*

Definition 1.32 (Centre). *The centre of a group G is defined as the subset of elements that commute with all elements of G , i.e.*

$$Z(G) := \{x \in G \mid xg = gx, \forall g \in G\}.$$

Theorem 1.33. *The centre of a group G is a subgroup, i.e. $Z(G) \leq G$.*

We will soon see an elegant way to understand why the normalizer and the centre are subgroups.

1.6.2 Stabilizers and Kernels

Now, we define some classes of subgroups that are defined with respect to a *group action* of G . In particular, again, we consider some non-empty subset $A \subseteq G$. We will also see that the preceding subgroups (i.e. centralizers, normalizers, and centre) can be defined as special cases of the constructions we will see now.

Recall that a group action is a mapping $G \times A \rightarrow A$ such that the mapping is associative and $1 \cdot a = a$.

Definition 1.34 (Stabilizer). *If G is a group acting on a set A , and $a \in A$ is some fixed element, then the stabilizer of a in G is the set*

$$G_a = \{g \in G \mid g \cdot a = a\}$$

where $g \cdot a$ is the group action of G on A .

Theorem 1.35. *The stabilizer is a subgroup, i.e. for any $a \in A \subseteq A$, $G_a \leq G$*

Definition 1.36 (Kernel of Group Action). *If the group G is acting on a set A , then the kernel of the group action is defined as*

$$\{g \in G \mid g \cdot a = a, \forall a \in A\}.$$

Theorem 1.37. *The kernel of a group action of G on a set A is a subgroup of G .*

Now, we finally come to the main observation where we note that several of these constructions can be unified by the notion of **conjugation**. Let $X = \mathcal{P}(G)$ be the set of all subsets of the group G . Now, suppose G acts on X by conjugation, i.e. for any $g \in G$ and $M \in X = \mathcal{P}(G)$, we say $g \cdot M = gMg^{-1} = \{gmg^{-1} \mid m \in M\}$. Then, we can note the following:

1. the normalizer, $N_G(A)$, is the stabilizer of A in G , i.e. $N_G(A) = G_A$ where $A \in \mathcal{P}(G)$
2. if the normalizer, $N_G(A)$ acts on the set A by conjugation, i.e. $g \cdot a = gag^{-1}$, $\forall a \in A$ and $g \cdot a \in A$ since $g \in N_G(A)$, then $C_G(A)$ is the kernel of this action. In other words, **the centralizer is the kernel of the action of $N_G(A)$ on A by conjugation**. As such,

$$C_G(A) \leq N_G(A).$$

3. the center of a group G , $Z(G)$, is the kernel of G acting on G by conjugation. As such,

$$Z(G) \leq G.$$

1.7. CYCLIC GROUPS

Definition 1.38 (Cyclic Groups). A **cyclic group** is a group H that is generated by a single element, i.e.

$$H = \{x^n \mid n \in \mathbb{Z}\}.$$

Note that n iterates over all integers, including negative integers. In other words, we product x with both itself and its inverse.

Here, we wrote the group operation as multiplication as just a matter of notation; we could have also written the group operation as addition, which would lead us to write $H = \{nx \mid n \in \mathbb{Z}\}$. In either case, we denote a cyclic group as $H = \langle x \rangle$ and say that H is a group that is generated by x .

Observation. The generator in a cyclic group need not be unique. For example, if $H = \langle x \rangle$, we could also write $H = \langle x^{-1} \rangle$. This is because:

- In multiplicative notation where $H = \{x^n \mid n \in \mathbb{Z}\}$, any x^n can be written as $(x^{-1})^{-n}$, so $H = \{x^{-1} \mid n \in \mathbb{Z}\}$.
- In additive notation where $H = \{nx \mid n \in \mathbb{Z}\}$, any nx can be written as $(-n)(-x)$, so $H = \{n(-x) \mid n \in \mathbb{Z}\}$.

Examples 1.39. We take a look at some examples of cyclic groups. They often appear as subgroups of larger groups that may not be cyclic themselves.

Consider the dihedral group: $G = D_{2n}$. Now, let H be the group of all rotations. Then, $H = \langle r \rangle = \{r^m \mid m \in \mathbb{Z}\}$. Note that $|H| = n$ since $r^n = 1 = r^0$. Also, $r^{-1} = r^{n-1}$.

The group of integers is cyclic i.e. $\mathbb{Z} = \langle 1 \rangle$.

1.7.1 Cardinality of cyclic group

We immediately note that there is a simple way of figuring out the cardinality of a cyclic group via its generator:

Proposition 1.40. Suppose that H is a cyclic group, i.e. $H = \langle x \rangle$. Then, $|H| = |x|$ (note that $|H|$ and $|x|$ are allowed to be infinite).

- if $|H| = n < \infty$, then $x^n = 1$ and the distinct elements of H are $\{1, x, \dots, x^{n-1}\}$.
- if $|H| = \infty$, then $x^n \neq 1$ for any $n \neq 0$.

Proof. First, suppose $|x| = n$. For any arbitrary $x^k \in H$, we can write using the division algorithm that $k = nq + r$ where $0 \leq r < n$. Then, $x^k = x^{nq+r} = (x^n)^q \cdot x^r = x^r \in \{1, x, \dots, x^{n-1}\}$. One can easily show that $x^a \neq x^b$ is $a, b \in \{1, \dots, n-1\}$ and $a \neq b$: if not, then, $x^{b-a} = 1$ but we know that n is the smallest positive integer such that $x^n = 1$. The proof for the infinite order is simple: no two x^a and x^b are equal for $a \neq b$ since, otherwise, we have an n such that $x^n = 1$ where $0 < n < \infty$. $\square$

Lemma 1.41. Let $x \in G$ where G is a group and let $m, n \in \mathbb{Z}$. If $x^m = 1 = x^n$, then

$$x^d = 1, \quad \text{where } d = (m, n).$$

In particular, if $x^m = 1$, then $|x|$ divides m .

Now, we come to an important result:

Theorem 1.42 (Classification of Cyclic Groups). **Any two cyclic groups of the same order are isomorphic.**

1. Suppose the cyclic groups $\langle x \rangle$ and $\langle y \rangle$ are both of order $n \in \mathbb{Z}_{>0}$. Then, the following is an isomorphism: $\varphi(x^k) = y^k$.

2. Suppose $\langle x \rangle$ is a cyclic group, then $\varphi(k) = x^k$ is an isomorphism. This follows from the observation that if $\langle x \rangle$ is of order n , then $\langle x \rangle \cong \mathbb{Z}/n\mathbb{Z}$.
3. In particular, for any $n \in \mathbb{Z}_{>0}$, let Z_n be the cyclic group of order n , then

$$Z_n \cong \mathbb{Z}/n\mathbb{Z}.$$

Proof Sketch: First we show that φ is well-defined. Suppose $x^r = x^s$. Then, we show that $\varphi(x^r) = \varphi(x^s)$. Note that $x^{r-s} = 1$, so $n|(r-s)$ by our previous lemma. Write $(r-s) = nt$. Then, $\varphi(x^r) = \varphi(x^{tn+s}) = y^{tn+s} = y^s = \varphi(x^s)$.

1.8. FORMULAS FOR ORDERS AND FINDING GENERATORS

In this section, we try to find the generators of a cyclic group.

First, we try to calculate the order of arbitrary powers of x in an arbitrary group. **Note:** the following result applies to an arbitrary group, not necessarily a cyclic group.

Proposition 1.43. Let G be a group (not necessarily cyclic) and let $x \in G$. Let $a \in \mathbb{Z} - \{0\}$.

1. if $|x| = \infty$, then $|x^a| = \infty$
2. if $|x| = n < \infty$, then $|x^a| = \frac{n}{(n,a)}$
3. if $|x| = n < \infty$ and $a|n$, then $|x^a| = \frac{n}{a}$.

Now, we use this to understand which elements in a cyclic group are generators of the group.

Proposition 1.44. Let H be a cyclic group, i.e. $H = \langle x \rangle$. Then,

1. If $|x| = \infty$, then $H = \langle x^a \rangle$ if and only if $a = 1$ or $a = -1$.
2. If $|x| = n < \infty$, then $H = \langle x^a \rangle$ if and only if $(a, n) = 1$.

Proof Sketch: we know that $\langle x^a \rangle$ generates a group of order $|x^a|$, so if $\langle x^a \rangle = \langle x \rangle$, we require that $|x^a| = |x|$. By 1.43, this is true if and only if $(a, n) = 1$.

Example 1.45. Consider $\mathbb{Z}/12\mathbb{Z}$. Then, we know that $\bar{1}$, $\bar{5}$, $\bar{7}$, and $\bar{11}$ are generators, since for each of these $(a, 12) = 1$.

Theorem 1.46. Let $H = \langle x \rangle$ be a cyclic group.

1. Every subgroup of H is cyclic. More precisely, if $K \leq H$, then either $K = \{1\}$ (i.e. the trivial group) or $K = \langle x^d \rangle$, where d is the smallest positive integer such that $x^d \in K$.
2. If $|H| = \infty$, then for any distinct nonnegative integers a and b , $\langle x^a \rangle \neq \langle x^b \rangle$. Furthermore, for every integer m , $\langle x^m \rangle = \langle x^{|m|} \rangle$, where $|m|$ denotes the absolute value of m . **In other words, the nontrivial subgroups of H correspond bijectively with the integers $1, 2, 3, \dots$**
3. If $|H| = n < \infty$, then for each positive integer a dividing n there is a unique subgroup of H of order a . This subgroup is the cyclic group $\langle x^d \rangle$, where $d = \frac{n}{a}$.

Furthermore, for every integer m , $\langle x^m \rangle = \langle x^{(n,m)} \rangle$, so that **the subgroups of H correspond bijectively with the positive divisors of n .**

1.9. SUBGROUPS GENERATED BY SUBSETS

The next class of subgroups we will look at are subgroups that are generated by subsets of a group. Let G be a group and let A be a subset of a group. We will define the subgroup generated by A to be the intersection of all subgroups of G that contain A . Naturally, this raises the question: is this intersection guaranteed to be a group? We prove this using the following lemma:

Lemma 1.47. *Let $\mathcal{C} \neq \emptyset$ be a collection of subgroups of a group G . Then, the intersection of all elements of $\mathcal{C}$ is a subgroup of G .*

Proof. Define $K = \bigcap_{X \in \mathcal{C}} X$. Given $1_G \in X$ (since X is defined to be a subgroup for each $X \in \mathcal{C}$), $1_G \in K$. If $x \in K$, then $x \in X$ for all $X \in \mathcal{C}$, which implies $x^{-1} \in X$ for all $X \in \mathcal{C}$, which implies $x^{-1} \in K$. This logic can be used to show that if $x, y \in K$, then $xy^{-1} \in K$. $\square$

Definition 1.48 (Subgroup generated by a Subset). *Let G be a group and let A be a subset of G . Then, the subgroup of G generated by A is defined as the intersection of all subgroups of G containing A , i.e.*

$$\langle A \rangle := \bigcap_{A \subseteq H, H \leq G} H. \quad (1)$$

In particular, for notational clarity, we set the following:

1. If $A = \{a_1, \dots, a_n\}$ is a finite set, we write $\langle a_1, \dots, a_n \rangle = \langle A \rangle$.
2. If A and B are two subsets, we write $\langle A, B \rangle = \langle A \cup B \rangle$.

Observation. Note that $\langle A \rangle$ is a subgroup by our last lemma if we let $\mathcal{C} = \{H \mid H \leq G, A \subseteq H\}$.

So far, we have only said that this definition gives us a subgroup that is generated by a subset. However, we do not yet know what the elements of this subgroup look like. We remedy this in the next result.

Definition/Proposition 1.49. *The subgroup of G that is generated by the subset $A \subseteq G$ can be written as:*

$$\langle A \rangle = \{a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_t^{\epsilon_t} \mid a_i \in A, \epsilon \in \{\pm 1\}, t \in \mathbb{Z}\} =: \bar{A}. \quad (2)$$

Proof. It is easy to show that $\bar{A}$ is a subgroup. To show that $\langle A \rangle = \bar{A}$, we first note that $\langle A \rangle \subseteq \bar{A}$ since each $a \in A$ can be written as $a^1 \in \bar{A}$. To show the reverse direction, we simply note that products and inverses belong to the group, so $\bar{A} \subseteq \langle A \rangle$. $\square$

1.10. QUOTIENT GROUPS

Definition 1.50 (Fibers). *Given $\varphi : G \rightarrow H$ is a group homomorphism, the fibers of φ are the sets of elements of G that φ maps to single elements. In other words, the fiber of φ above $h \in H$, denoted by X_h , is defined as:*

$$X_h = \{g \in G \mid \varphi(g) = h\}.$$

Observation. Note that there is a natural operation we can do on fibers themselves because of the properties of homomorphisms. Let X_h and $X_{h'}$ be two fibers. Then, $X_h X_{h'}$ is the fiber $X_{hh'}$. This is because for any $\alpha \in X_h, \beta \in X_{h'}$, $\varphi(\alpha\beta) = \varphi(\alpha)\varphi(\beta) = hh'$.

This will allow us to define the quotient group as the group of fibers of a homomorphism.

Definition 1.51 (Kernel and image of a homomorphism). *Let $\varphi : G \rightarrow H$ be a group homomorphism. Then, the kernel of φ is*

$$\ker \varphi := \{g \in G \mid \varphi(g) = 1_H\}.$$

On the other hand, the image of the homomorphism is defined as

$$\text{im } \varphi := \{h \in H \mid \varphi(g) = h \text{ for some } g \in G\}.$$

We look at some immediate properties of kernels and images:

Proposition 1.52. Let $\varphi : G \rightarrow H$ be a group homomorphism. Then,

1. $\varphi(1_G) = 1_H$, $\varphi(g^{-1}) = \varphi(g)^{-1}$ and $\varphi(g^n) = \varphi(g)^n$
2. $\ker \varphi$ and $\text{im } \varphi$ are subgroups of G and H respectively, i.e. $\ker \varphi \leq G$ and $\text{im } \varphi \leq H$.

Now, we define the quotient group:

Definition 1.53 (Quotient Group). Let $\varphi : G \rightarrow H$ be a group homomorphism with $K = \ker \varphi$. Then, the quotient group, denoted by G/K , is the group whose elements are fibers X_h for each $h \in H$ and the operation is defined by $X_h X_{h'} = X_{hh'}$. Note that K is also an element in G/K ; concretely $K = X_{1_H}$ and K is the identity of the quotient group G/K .

The odd thing about this definition is that X_h is very opaque. What we will now see is that each fiber is essentially a translate of the kernel K via some representative:

Proposition 1.54. Let $\varphi : G \rightarrow H$ be a group homomorphism and let $K := \ker \varphi$ be its kernel. Then, $X_h \in G/K$ can be written as, for any $g' \in X_h$,

$$X_h = g'K := \{g'k \mid k \in K\}$$

and

$$X_h = Kg' := \{kg' \mid k \in K\}.$$

Proof. It is easy to see that $g'K \subseteq X_h$ because $\varphi(g'k) = h$ for any $k \in K$. To see the reverse inclusion, let $g \in X_h$, i.e. $\varphi(g) = h$. Now, define $k = g'^{-1}g$. Then, $\varphi(k) = \varphi(g'^{-1}g) = \varphi(g')^{-1}\varphi(g) = h^{-1}h = 1$. So, $k \in K$. Therefore, $g = g'k \in g'K$. $\square$

We used a particularly general construction in this definition that is worth noting separately:

Definition 1.55 (Left coset/Right coset and Representatives). For a subgroup N of G and some element $g \in G$, we will use the notation $gN = \{gn \mid n \in N\}$ which is called a **left coset** of N in G . Similarly, $Ng = \{ng \mid n \in N\}$ which is called a **right coset** of N in G . Any element of a coset is called a **representative** of the coset. In particular, $g = g1 = 1g$ itself is a representative of the coset gN or Ng .

Note that if G is an additive group, we will write $g+N$ to be the left coset of N in G with representative g (and the equivalent statement for right cosets is true).

Observation. The quotient group G/K consists of elements that are the left cosets of $K = \ker \varphi$ in G . In other words, the elements of G/K look like gK where g is a representative of a fiber. Furthermore, since different representatives define the same fiber, the choice of the representative does not matter – we will formalize this now:

Proposition 1.56. Let G be a group and let $K = \ker \varphi$ for some group homomorphism φ . Then, consider the set of left cosets of K in G and define the following operation on them: $uK \circ vK = (uv)K$. Then,

1. This operation defined a group, called the quotient group, denoted by G/K
2. The operation is well-defined, i.e. if $u_1 \in uK$ and $v_1 \in vK$, then $u_1v_1 \in uvK$ and $u_1v_1K = uvK$. In other words, the group operation does not depend on the choice of representative.
3. The statements are true after replacing left cosets with right cosets.

In light of all this, we write the following as the definition of a quotient group:

Definition 1.57 (Quotient Group). Let $\varphi : G \rightarrow H$ be a group homomorphism and let $K = \ker \varphi$. Then, the quotient group G/K is defined as the cosets of the kernel $K = \ker \varphi$ in G , i.e.

$$G/K := \{\bar{g} := gK \mid g \in G\}$$

with the group operation being $\bar{g} \cdot \bar{g}' = \overline{gg'}$, i.e. $gK \cdot g'K = (gg')K$. Note that $\bar{g} = \bar{g}'$ if $g' \in gK$ or $g \in g'K$.

Example 1.58. Let $\varphi : G \rightarrow H$ be an isomorphism. Then, the kernel is just 1. In particular, the fibres of φ are singleton subsets of G (since φ is injective and surjective), so $G/1 \cong G$.

Example 1.59. Consider the homomorphism $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$. This has the kernel $K = n\mathbb{Z}$. Then, the cosets of K in $\mathbb{Z}$ are $a + K = a + n\mathbb{Z}$ for each $a \in \mathbb{Z}$.

Example 1.60. let $G = \mathbb{R}^2$ and let $H = \mathbb{R}$. Define $\varphi : G \rightarrow H$ by $\varphi((x, y)) = x$. Then, one can show that φ is a homomorphism and its kernel is $\ker \varphi = \{(0, y) \mid y \in \mathbb{R}\}$. Then, the quotient group G/K consists of the cosets $\overline{(x, 0)} = (x, 0) + \ker \varphi$ for each $x \in \mathbb{R}$.

1.10.1 Important properties of cosets and representatives

Some of what we have seen so far can be generalized. For example, the fact that a coset can be written using any representative and the choice of the representative does not make a material difference in defining the coset can be generalized to cosets of other subgroups as well alongside cosets of a kernel. On the other hand, we will see that the group operation in the quotient group, i.e. $uK \cdot vK = uvK$, is only defined if K satisfies some properties.

Proposition 1.61. *Let N be any subgroup of G , i.e. $N \leq G$. Then, the set of left cosets of N (sets of the form gN for $g \in G$) partition G , i.e. all elements of G belong to some coset of N in G . In particular, for all $u, v \in G$, $uN = vN$ if and only if $v^{-1}u \in N$. In particular, $uN = vN$ if and only if u and v are representatives of the same coset.*

Proof. Since $1 \in N$ (as N is a subgroup), each element of G can be written as $g = g \cdot 1 \in gN$. As such, g is a representative of the coset gN and, therefore, $G = \cup_{g \in G} gN$. We must now show that distinct cosets have no intersection. In other words, if $uN \cap vN \neq \emptyset$, then $uN = vN$. Let $x \in uN \cap vN$. Then, $x = un = vm$ for some $m, n \in N$. now, $u = vmn^{-1} = vm_1$ where $m_1 := mn^{-1} \in N$. Then, $uN = \{ut \mid t \in N\} = \{vm_1t \mid t \in N\} = vm_1N \subseteq vN$. Similarly, $vN \subseteq uN$, which gives us $uN = vN$.

Now, we prove the second statement. We know that $uN = vN$ if and only if $u \in vN$ if and only if $u = vn$ for some $n \in N$ if and only if $v^{-1}u \in N$.

Now, we prove the last statement. Again, $uN = vN$ if and only if $v \in uN$ which is equivalent to saying $v = v \cdot 1_G$ is a representative of uN . Similarly, $uN = vN$ if and only if $u \in vN$ which is equivalent to saying u is a representative of vN . Since $uN = vN$, therefore, u and v are representatives of the same coset. $\square$

Since this makes remembering almost everything so far quite simple, it is worth writing it again: **Two cosets are the same, i.e. $uN = vN$, if and only if u and v are representatives of the same coset. Also, two cosets are the same, i.e. $uN = vN$, if and only if $u \in vN$.**

Note that, in additive notation, we would say that two cosets are the same, i.e. $uN = vN$, if and only if $u \in v + N$

Now, we come to the second part: what are the requirements on the subgroup N that make $uN \circ vN = (uv)N$ a well-defined group operation?

Proposition 1.62. *Let G be a group and let N be a subgroup, $N \leq G$.*

1. *The operation $uN \circ vN = (uv)N$ is well-defined if and only if $gng^{-1} \in N$ for all $g \in G, n \in N$ (as we will soon see - this means, this operation is well-defined if and only if N is a normal subgroup).*
2. *If this condition is satisfied, then the set of left cosets of N in G , i.e. $\{gN \mid g \in G\}$, is a group with the group operation $gN \circ g'N = (gg')N$. In particular, the identity is the coset $1N$ and the inverse of gN is $g^{-1}N$.*

Proof. We sketch the proof of the first part. Assume that the operation is well-defined. Then, suppose $u, u_1 \in uN$ and $v, v_1 \in vN$. We know that $uN \circ vN = (uv)N$. But choosing different representatives of uN and vN , we get that $u_1N \circ v_1N = (u_1v_1)N$ must be the same as $(uv)N$, so $(uv)N = (u_1v_1)N$. Now, let $u = 1, u_1 = n$ for some $n \in N$ (clearly, then $u, u_1 \in uN$) and let $v = v_1 = g^{-1}$ for some fixed $g \in G$. Then, $1g^{-1}N = ng^{-1}N$ which implies $g^{-1}N = ng^{-1}N$. Clearly, $ng^{-1} \in ng^{-1}N$, so $ng^{-1} \in g^{-1}N$, so $ng^{-1} = g^{-1}n'$ for some $n' \in N$, which gives us that $gng^{-1} \in n' \in N$. Conversely, suppose $gng^{-1} \in N$

for all $g \in G, n \in N$. Again, let $u, u_1 \in uN$ and $v, v_1 \in vN$. Write $u_1 = un$ and $v_1 = vm$ for some $n, m \in N$. We can show that $u_1v_1 = (uv)(n_1m)$ where $n_1 = v^{-1}nv$ which is in N by assumption. Since N is closed under products, $n_1m \in N$, so $u_1v_1 = (uv)n_2$ for some $n_2 \in N$ (precisely $n_2 = n_1m \in N$). Since $u_1v_1 \in uvN$, $u_1v_1N = uvN$. As such, the operation is well-defined. $\square$

Subgroups that satisfy these properties are extremely important in the study of groups and, as such, are given a separate name.

Definition 1.63 (Normal Groups and conjugates). *Let G be a group and let N be a subgroup of G . The element gng^{-1} is called the conjugate of $n \in N$ by $g \in G$. The set $gNg^{-1} = \{gng^{-1} \mid n \in N\}$ is called the conjugate of N by g .*

*In particular, the element g is said to normalize N if $gNg^{-1} = N$. A subgroup N of a group G is called a **normal subgroup** if $gNg^{-1} = N$ for all $g \in G$, i.e. every element of G normalizes N . If N is a normal subgroup of G , we write $N \trianglelefteq G$.*

Theorem 1.64. *Let N be a subgroup of the group G . Then, the following are equivalent:*

1. $N \trianglelefteq G$
2. $N_G(N) = G$, i.e. the normalizer of N in G is G itself
3. $gN = Ng$ for all $g \in G$
4. The set of left cosets are a group under the operation $uN \circ vN = (uv)N$
5. $gNg^{-1} \subseteq N$ for all $g \in G$.

Note that the kernel of a homomorphism satisfies both the requirements in lemma 1.62. One can ask whether all subgroups N that satisfy these properties can be shown to be the kernel of some homomorphisms. This is answered in the next proposition:

Proposition 1.65. *A subgroup N of the group G is normal if and only if it is the kernel of some homomorphism.*

Proof. The kernel of a homomorphism is normal as we have already shown. We need to now show that if $N \trianglelefteq G$, then there exists a homomorphism whose kernel is N . Define $\varphi : G \rightarrow G/N$ by $\varphi(g) = gN$ for all $g \in G$. Note that this is a homomorphism as $\varphi(g_1g_2) = (g_1g_2)N = g_1N \cdot g_2N = \varphi(g_1)\varphi(g_2)$. Now, it is easy to check that $\ker \varphi = \{g \in G \mid \varphi(g) = 1N\} = \{g \in G \mid gN = 1N\} = \{g \in G \mid g \in N\} = N$. $\square$

Definition 1.66. *Let $N \trianglelefteq G$ be a normal subgroup. The homomorphism $\pi : G \rightarrow G/N$ defined by $\pi(g) = gN$ is called the **normal projection/homomorphism** of G onto G/N . If $\bar{H} \leq G/N$ is a subgroup of G/N , the complete preimage of $\bar{H}$ in G is the preimage of $\bar{H}$ under the normal projection homomorphism.*

Example 1.67. For any group G , the subgroups 1 and G are normal in G . In particular, $G/1 \cong G$ and $G/G \cong 1$.

Example 1.68. Let G be an abelian group, then any subgroup N of G is a normal subgroup since $gng^{-1} = gg^{-1}n = n \in N$ for all $g \in G, n \in N$.

Example 1.69. Let $G = Z_k$ be the cyclic group of order k . Let x be a generator of G and let $N \leq G$. Then, $N = \langle x^d \rangle$ where d is the smallest integer such that $x^d \in N$. Then,

$$G/N = \{gN \mid g \in G\} = \{x^\alpha N \mid \alpha \in \mathbb{Z}\}.$$

Now, since $x^\alpha N = (xN)^\alpha$, $G/N = \langle xN \rangle$, i.e. G/N is cyclic and its generator is xN . Furthermore, $|G/N| = \frac{|G|}{|N|}$. **In other words, the quotient groups of cyclic groups are cyclic.**

We now prove the following theorem, known as Lagrange's Theorem, which allows us to compute the order of a quotient group.

Theorem 1.70 (Lagrange's Theorem). *If G is a finite group and H is a subgroup of G , then the order of H divides the order of G and the number of left cosets of H in G equals $\frac{|G|}{|H|}$.*

Proof. Let $|H| = n$ and let the number of left cosets of H in G equal k . Now, we claim that for any fixed $g \in G$, $|gH| = n$. This can be seen by noting that the map $H \rightarrow gH$ by $h \rightarrow gh$ is (a) surjective by definition and (b) injective because (if $gh_1 = gh_2$, then $h_1 = h_2$), and so $|gH| = |H| = n$.

Now, we have proven previously (see Proposition 1.61) the left cosets of H in G partition G , so $|G| = |\text{no. of left cosets}| \cdot |\text{cardinality of each left coset}| = kn$. As such, number of left cosets of H in G is $k = \frac{|G|}{n} = \frac{|G|}{|H|}$. $\square$

One point to note and remember from this proof is the idea that $|H| = |gH|$ and that $H \rightarrow gH$ is a bijection.

Definition 1.71 (Index of a subgroup). *Let G be a group and let H be a subgroup of G . Then, the index of H in G is the number of left cosets of H in G . We denote the index by $|G : H|$.*

Note that if G is a finite group, then by Lagrange's theorem, $|G : H| = \frac{|G|}{|H|}$. However, if G is an infinite group, then this is not true. In particular, a subgroup of an infinite group can have finite index or infinite index. As an example, consider the group $\mathbb{Z}$. Then, the trivial subgroup $\{0\}$ has order 1 but has infinite index. On the other hand, the subgroup $\langle n \rangle$ is of infinite order but has index n (assuming $n > 0$).

Corollary 1.72. *let G be a group. We have the following consequences of Lagrange's theorem:*

1. *If G is a finite group and let $x \in G$. Then, $|x|$ divides the order of G . In particular, $x^{|G|} = 1$ for all $x \in G$.*
2. *If G is a group of order p where p is a prime integer. Then, G is cyclic, hence $G \cong Z_p$. (Recall that Z_p is the cyclic group of order p and $Z_p \cong \mathbb{Z}/p\mathbb{Z}$)*

Example 1.73. Let $H = \langle (1, 2, 3) \rangle \leq S_3$ and let $G = S_3$. We now show that $H \trianglelefteq S_3$ using Lagrange's theorem. First, one can show that $H \leq N_G(H) \leq G$. Now, by Lagrange's theorem, the order of H divides the order of $N_G(H)$ which divides the order of S_3 , which is 6. Now, since G has order 6 and H has order 3, the only possibilities for $N_G(H)$ are that $N_G(H)$ is either the same as H or the same as $G = S_3$. Now, one can show that $(1, 2)$ conjugates $(1, 2, 3)$ to $(1, 2, 3)^{-1}$. So, $(1, 2)$ conjugates a generator of H to another generator of H , which allows us to conclude that $(1, 2) \in N_G(H)$. Since $(1, 2) \notin H$, so, $N_G(H) \neq H$ and $N_G(H) = G$, which allows us to conclude that $H \trianglelefteq S_3$.

Definition 1.74 (Simple Groups). *Simple groups are groups whose only normal subgroups are 1 and G .*

Theorem 1.75 (Cauchy's Theorem). *If G is a finite group and p is a prime dividing $|G|$, then G has an element of order p .*

Theorem 1.76 (Sylow). *If G is a finite group of order $p^\alpha m$, where p is a prime and p does not divide m , then G has a subgroup of order p^α .*

Proposition 1.77. *If H and K are finite subgroups of a group, then*

$$|HK| = \frac{|H||K|}{|H \cap K|}$$

where $HK = \{hk \mid h \in H, k \in K\}$.

Proposition 1.78. *If H and K are subgroups of a group, HK is a subgroup if and only if $HK = KH$.*

Corollary 1.79. *If H and K are subgroups of G , and $H \leq N_G(K)$, then $HK \leq G$. In particular, if $K \trianglelefteq G$, then $HK \leq G$ for any $H \leq G$.*

1.11. GROUP ISOMORPHISM THEOREMS

Theorem 1.80 (First Isomorphism Theorem). *Let $\varphi : G \rightarrow H$ be a group homomorphism. Then, $\ker \varphi \trianglelefteq G$ and $G/\ker \varphi \cong \varphi(G)$.*

Proof. Let $x \in \ker \varphi$. Then, for any $g \in G$, $\varphi(gxg^{-1}) = \varphi(g)\varphi(x)\varphi(g)^{-1} = \varphi(g)1_H\varphi(g)^{-1} = \varphi(g)\varphi(g)^{-1} = 1_H$, so $gxg^{-1} \in \ker \varphi$. We already know that $\ker \varphi$ is a subgroup, so $\ker \varphi \trianglelefteq G$. To prove the second statement, define $\Psi : G/\ker \varphi \rightarrow \varphi(G)$ by $\Psi(g\ker \varphi) = \varphi(g)$. Now, with $K = \ker \varphi$,

- Ψ is well-defined. Suppose $gK = hK$. Then, $h^{-1}g \in K$. Then, $\varphi(h^{-1}g) = 1_H$. But $\varphi(h^{-1}g) = \varphi(h)^{-1}\varphi(g) = 1_H$, which implies $\varphi(h) = \varphi(g)$. So, $\Psi(g) = \Psi(h)$.
- Ψ is a homomorphism since $\Psi(gK \circ hK) = \Psi(ghK) = \varphi(gh) = \varphi(g)\varphi(h) = \Psi(gK)\Psi(hK)$.
- Ψ is injective. This is because $\Psi(gK) = \Psi(hK) \implies \varphi(g) = \varphi(h)$. But then, $\varphi(h)^{-1}\varphi(g) = 1_H$, which implies $\varphi(h^{-1}g) = 1_H$. So $h^{-1}g \in K$, so $gK = hK$.
- Ψ is surjective by definition of the morphism.

□

Corollary 1.81. *Let $\varphi : G \rightarrow H$ be a group homomorphism. Then, φ is injective if and only if $\ker \varphi = 1$. Furthermore, $|G/\ker \varphi| = |\varphi(G)|$.*

Proof. If φ is injective, then, $\ker \varphi = 1_G$ by definition of an injective map. On the other hand, if $\ker \varphi = 1_G$, then suppose $\varphi(x) = \varphi(y)$. Then, $\varphi(y)^{-1}\varphi(x) = \varphi(y^{-1}x) = 1_H$. Since $y^{-1}x$ is in the kernel and the kernel is simply 1_G , we have that $y^{-1}x = 1_G$, so $x = y$. To prove the second statement, we simply use the first isomorphism theorem (1.80) and Lagrange's theorem (1.70). □

Theorem 1.82 (Second/Diamond Isomorphism Theorem). *Let G be a group and let A and B be subgroups of G . Assume that $A \leq N_G(B)$. Then, $AB \leq G$, $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$, and $AB/B \cong A/A \cap B$.*

Proof. The first statement is the same as 1.79. To show $B \trianglelefteq AB$, we note that $A \leq N_G(B)$ by hypothesis and $B \leq N_G(B)$ trivially, so $AB \leq N_G(B)$. This means, all elements of AB normalize B . As such, B is a normal subgroup of AB considered as a group.

Now, given $B \trianglelefteq AB$, we can define the quotient group AB/B and that allows us to construct the map $\varphi : A \rightarrow AB/B$ by $\varphi(a) = aB$. It is straightforward to check that this is well-defined and a homomorphism. From the definition, clearly, φ is surjective. On the other hand, the kernel of φ is $A \cap B$. As such, $\ker \varphi = A \cap B$ which, by the first isomorphism theorem, is a normal subgroup of A . The first isomorphism theorem also allows us to conclude the last statement. □

Theorem 1.83 (Third Isomorphism Theorem). *Let G be a group. Let $H \trianglelefteq G$ and $K \trianglelefteq G$, and let $H \leq K$. Then,*

$$K/H \trianglelefteq G/H$$

and $(G/H)/(K/H) \cong G/K$, which can be equivalently written with the bar notation (i.e. $\bar{J} = J/H$) as:

$$\bar{G}/\bar{K} \cong G/K.$$

Proof. The fact that $K/H \trianglelefteq G/H$ follows from the fact that $K \trianglelefteq G$ and the group operation in quotient group, i.e. $uK \circ vK = (uv)K$. To prove the second statement, construct the following map: $\varphi : G/H \rightarrow G/K$ which sends gH to gK . First, φ is well-defined: if $g_1H = g_2H$, then $g_1 = g_2h$ for some $h \in H$, and since $H \leq K$, $g_1K = g_2K$. So $\varphi(g_1H) = \varphi(g_2H)$. On the other hand, φ is, clearly, a surjective homomorphism. Finally, consider its kernel:

$$\begin{aligned} \ker \varphi &= \{gH \in G/H \mid \varphi(gH) = 1K\} \\ &= \{gH \in G/H \mid gK = 1K\} \\ &= \{gH \in G/H \mid g \in K\} \\ &= K/H. \end{aligned}$$

Using the first isomorphism theorem, we get the final statement in the theorem. □

Finally, we come to the fourth isomorphism theorem, which is arguably more important than the last two:

Theorem 1.84 (Fourth or Lattice Isomorphism Theorem / Correspondence Theorem). *Let G be a group and let $N \trianglelefteq G$. Then, there is the following bijection:*

$$\{ \text{Subgroups } A \leq G \text{ s.t. } N \subseteq A \} \leftrightarrow \{ \text{Subgroups } \bar{A} = A/N \leq G/N \}.$$

In particular, any subgroup of G/N is of the form A/N for some subgroup A of G containing N . Furthermore, this bijection is inclusion-preserving (i.e. if $H_1 \subseteq H_2$, then $H_1/N \subseteq H_2/N$).

In particular, the bijection has the following properties: for all $A, B \leq G$ with $N \leq A$ and $N \leq B$:

1. $A \leq B$ if and only if $\bar{A} \leq \bar{B}$ (recall that $\bar{A} = A/N$)
2. if $A \leq B$, then $|B : A| = |\bar{B} : \bar{A}|$
3. $\langle \bar{A}, \bar{B} \rangle = \overline{\langle A, B \rangle}$
4. $\overline{A \cap B} = \bar{A} \cap \bar{B}$
5. $A \trianglelefteq G$ if and only if $\bar{A} \trianglelefteq \bar{G}$.

Proof Sketch: The mapping is straightforward; it is the normal projection which takes $\pi(x) = xN$ for any $x \in G$. We first show that the map is well-defined. Let $A \leq G$ with $N \subseteq A$. Then, $\pi(A) = A/N$ is a subgroup since homomorphisms map subgroups to subgroups. Conversely, if A/N is a subgroup of G/N , then its preimage, $\pi^{-1}(A/N) = \{k \in G \mid kN \in A/N\} = A$. In particular, A must contain N because π maps N to $1N \in A/N$. It is straightforward to show that these maps are inverses of each other and preserves inclusion. While (1) to (4) are straightforward, (5) is quite useful. Suppose $A \trianglelefteq G$. Now, consider A/N . For any $xN \cdot aN \cdot (xN)^{-1} = Nx \cdot a \cdot x^{-1}N^{-1} = N \cdot a \cdot N^{-1} = aNN^{-1} = aN$, so $A/N \trianglelefteq G/N$. Converse is proven similarly.

Observation. *In all the theorems and examples, we have constructed homomorphisms on quotient groups, i.e. homomorphisms of the form $\varphi : G/N \rightarrow H$. In some of these cases, the value of φ on the coset gN is given in terms of the representative g alone. We then had to prove that φ is well-defined (i.e. the value of φ on the coset was independent of the choice of g). Now, we can get a homomorphism Φ from G to H that makes the following diagram commute:*

$$\begin{array}{ccc} G & \xrightarrow{\pi} & G/N \\ & \searrow \Phi & \downarrow \varphi \\ & & H \end{array}$$

In terms of these constructions, we have that

$$\varphi \text{ is well-defined on } G/N \text{ if and only if } N \leq \ker \Phi.$$

1.12. GROUP ACTIONS

Recall: a **group action** of a group G on a set A is a mapping $G \times A \rightarrow A$ such that:

1. $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a$ for all $g_1, g_2 \in G, a \in A$
2. $1_G \cdot a = a, \forall a \in A$.

With this, for each $g \in G$, we can define a *permutation* of A by

$$\sigma_g : A \rightarrow A, \quad \sigma_g(a) = g \cdot a.$$

We can also define a *permutation representation* associated to a given group action by the following homomorphism:

$$\varphi : G \rightarrow S_A, \quad \varphi(g) = \sigma_g.$$

Furthermore, we defined the following:

1. The kernel of a group action of G on A is the set $\{g \in G \mid g \cdot a = a, \forall a \in A\}$. The kernel of the group action is the same as the kernel of the corresponding permutation representation. Therefore, the kernel of a group action is a normal subgroup of G .
2. For each $a \in A$, we have the stabilizer of a in G which is the set $G_a := \{g \in G \mid g \cdot a = a\}$.
3. The group action is called faithful if the kernel is the identity 1_G .

Note:

1. Two group elements g and g' induce the same permutation on A (i.e. $\sigma_g = \sigma_{g'}$) if and only if they are in the same fibre of the permutation representation φ . This happens if and only if they are in the same coset of the kernel.
2. An action of G on A is a faithful action of $G/\ker \varphi$ on A .

Example 1.85. Consider the action of S_n on $A = \{1, 2, \dots, n\}$ (where $n \in \mathbb{Z}_{>0}$) by $\sigma \cdot i = \sigma(i)$. In this case, the permutation representation is the identity map, i.e. $\varphi : S_A \rightarrow S_A$ by $\varphi(\sigma) = \sigma$. The action is faithful and, for each $i \in \{1, \dots, n\}$, the stabilizer G_i is isomorphic to S_{n-1} .

We saw that given a group action of G on A , we can define a permutation representation, which is a homomorphism $\varphi : G \rightarrow S_A$. This can be reversed: if $\varphi : G \rightarrow S_A$ is a group homomorphism, we can define a group action of G on A by $g \cdot a = \varphi(g)(a)$. This is summarized in the following:

Proposition 1.86. *For any group G and a non-empty set A , there is the following bijection:*

$$\{\text{group actions of } G \text{ on } A\} \leftrightarrow \text{Hom}(G, S_A).$$

Given this result, we can redefine permutation representations as the following:

Definition 1.87 (Permutation Representation). *If G is a group, then a permutation representation of G is a homomorphism*

$$\varphi : G \rightarrow S_A$$

where S_A is the symmetric group for some non-empty set A . We say that a given group action of G on A induces a permutation presentation of G .

Note that in the case where A is a finite set of n elements, we can also say that $S_A \cong S_n$.

Proposition 1.88. *Let G be a group acting on a set $A \neq \emptyset$. The relation on A defined by $a \sim b$ if and only if $a = g \cdot b$ for some $g \in G$ is an equivalence relation. For each $a \in A$, the number of elements in the equivalence class containing a is $|G : G_a|$, the index of the stabilizer of a .*

Proof. To show that this is an equivalence relation. First, $a \sim a$ since $a = 1_G \cdot a$ by definition of a group action, i.e. the relation is reflexive. Secondly, if $a \sim b$, i.e. $a = g \cdot b$ for some $g \in G$, then $g^{-1} \cdot a = g^{-1} \cdot (g \cdot b) = 1_G \cdot b = b$, so $b \sim a$, i.e. the relation is symmetric. Finally, if $a \sim b$ and $b \sim c$, then $a = g \cdot b$ and $b = g' \cdot c$ for some $g, g' \in G$. Then, $a = g \cdot b = g \cdot (g' \cdot c) = \tilde{g} \cdot c$ where $\tilde{g} = g \cdot g' \in G$, so $a \sim c$, i.e. the relation is transitive.

To prove the second statement, we construct a bijection between the left cosets of G_a in G and the elements of the equivalence class of a . Denote the equivalence class of a as

$$\mathcal{C}_a = \{g \cdot a \mid g \in G\}.$$

Now, suppose $b = g \cdot a \in \mathcal{C}_a$. Then, gG_a is a left coset of G_a in G . The map $b = g \cdot a \rightarrow gG_a$. Then, gG_a is a left coset of G_a in G . The map $b = g \cdot a \rightarrow gG_a$ is a map from $\mathcal{C}_a$ to the set of left cosets of G_a in G . This is surjective since for any $g \in G$, the element $g \cdot a$ is an element of $\mathcal{C}_a$. Since $g \cdot a = h \cdot a$ if and only if $h^{-1}g \in G_a$ if and only if $gG_a = hG_a$, the map is injective, so this is a bijection. $\square$

1.13. CYCLE DECOMPOSITIONS

Definition 1.89 (Orbits). *Let G be a group acting on $A \neq \emptyset$. Then, the equivalence class $\{g \cdot a \mid g \in G\}$ is called the orbit of G containing a .*

Proposition 1.90. *Every element of the symmetric group S_n has a unique cycle decomposition. In other words, every element can be uniquely expressed as a product of disjoint cycles.*

Proof. Let $A = \{1, \dots, n\}$ and let $\sigma \in S_A = S_n$. Now, consider $G = \langle \sigma \rangle$. Then, G acts on A and, so, it partitions A into unique set of disjoint sets, i.e. the orbits.

Now, consider a fixed orbit $\mathcal{O} = \{g \cdot x \mid g \in G\}$ of G containing $x \in \mathcal{O}$. Then, there exists the following bijection:

$$\{\text{left cosets of } G_x \text{ in } G\} \leftrightarrow \{\text{elements of } \mathcal{O}\}$$

(recall that $G_x = \{g \cdot x \mid g \in G \text{ s.t. } gx = x\}$, so in our case $G_x = \{\sigma^j \mid \sigma^j \in G \text{ s.t. } \sigma^j x = x\}$) by

$$\sigma^i G_x \leftrightarrow \sigma^i x.$$

Since G is cyclic, we know that $G_x \trianglelefteq G$ and G/G_x is cyclic of order d which is the smallest positive integer such that $\sigma^d \in G_x$. On the other hand, because of the bijection, $d = |G : G_x| = |\mathcal{O}|$. Then, there are d distinct cosets of G_x in G which can be written as $\{G_x, \sigma G_x, \sigma^2 G_x, \dots, \sigma^{d-1} G_x\}$. Using the bijection, the distinct elements of $\mathcal{O}$ are $\{x, \sigma x, \sigma^2 x, \dots, \sigma^{d-1} x\}$. Therefore, σ is a d -cycle. This allows us to conclude that each σ has a cycle decomposition.

Now, the orbits of $\langle \sigma \rangle$ are uniquely determined by σ . Within each orbit, $\mathcal{O}$, we can begin with any element as a representative — edifferent choices simply produces a different cyclic permutation of the original list. As such, the cycle decomposition is unique up to a rearrangement of the cycles and up to a cyclic permutation of the integers within each cycle. $\square$

Definition 1.91 (Permutation Groups). *Subgroups of symmetric groups are called permutation groups.*

For each subgroup G of S_n , the orbits of G will refer to its orbit on $\{1, 2, \dots, n\}$.

1.14. CAYLEY'S THEOREM AND THE GROUP ACTION OF LEFT MULTIPLICATION

Let G act on itself via left multiplication, i.e. $g \cdot a = ga, \forall g, a \in G$. Note that if G is written additively, then $ga = g + a$. Now, if G is finite, we can label its elements as $\{g_1, \dots, g_n\}$. Then, each $g \in G$ describes a permutation $\sigma_g \in S_n$ via

$$\sigma_g(i) = j \leftrightarrow gg_i = g_j.$$

This group action, by left multiplication, is both transitive and faithful, as we will see in the next proposition. The stabilizer of any point is the identity subgroup.

Now, let $H \leq G$ be a subgroup. Let

$$A = \{\text{left cosets of } H \text{ in } G\}.$$

Then, G also acts on A by $g \cdot aH = gaH, \forall g \in A, aH \in A$. Note that if $H = \{1\}$, then $aH = \{a\}$ and, so, this action is the same as G acting on itself via left multiplication. On the other hand, if H is of finite index m (i.e. there are m left cosets of H in G), we can label the cosets of H with the integers from 1 to m . We can, then, label the left cosets of H in G as $a_1H, a_2H, \dots, a_mH$. Then, for each $g \in G$, we have a permutation of $\sigma_g \in S_m$, by

$$\sigma_g(i) = j \leftrightarrow ga_iH = a_jH.$$

Proposition 1.92. *Let G be a group and let $H \leq G$ be a subgroup. Let G act on $A = \{\text{left cosets of } H \text{ in } G\}$. Let π_H be the permutation representation of this action:*

$$\pi_H : G \rightarrow S_A.$$

Then,

1. G acts transitively on A
2. the stabilizer in G of the point $1_H \in A$ is the subgroup H .
3. the kernel of the action, i.e. $\ker \pi_H$, is $\cap_{x \in G} xHx^{-1}$.
4. $\ker \pi_H$ is the largest normal subgroup of G contained in H .

Proof. To prove (1), we show that there is only one orbit of this group action. Let $aH, bH \in A$. Then, let $g = ba^{-1}$. Then, $g \cdot aH = bH$, so any two arbitrary elements fall into the same orbit.

To prove (2), we note that the stabilizer of 1_H is $\{g \in G \mid g \cdot 1_H = 1_H\} = \{g \in G \mid gH = H\} = H$.

To prove (3), we calculate as follows:

$$\begin{aligned}
 \ker \pi_H &= \{g \in G \mid g \cdot xH = xH, \forall x \in G\} \\
 &= \{g \in G \mid (x^{-1}gx)H = H, \forall x \in G\} \\
 &= \{g \in G \mid x^{-1}gx \in H, \forall x \in G\} \\
 &= \{g \in G \mid g \in xHx^{-1}, \forall x \in G\} \\
 &= \cap_{x \in G} xHx^{-1}.
 \end{aligned}$$

To prove (4), we know already that $\ker \pi_H \trianglelefteq G$ (by first isomorphism theorem) and $\ker \pi_H \trianglelefteq H$. To see the latter, let $x = 1$ and we get that $\ker \pi_H \subset H$. Now, since $\ker \pi_H \trianglelefteq G$, for any $h \in H$, $hkh^{-1} \in \ker \pi_H$ for any $k \in \ker \pi_H$, so $\ker \pi_H \trianglelefteq H$. Now, let $N \trianglelefteq G$ such that $N \subseteq H$. Then, $N = xNx^{-1} \leq xHx^{-1}$ for any $x \in G$. Then, $N \subseteq \cap_{x \in G} xHx^{-1} = \ker \pi_H$. $\square$

Theorem 1.93 (Cayley's Theorem). *Every group is isomorphic to a subgroup of some symmetric group. In particular, if G is a group of order n , then, G is isomorphic to a subgroup of S_n .*

Proof. Let $H = 1$. Let G act on itself via left multiplication, which can be seen as G acting on $A = \{\text{left cosets of } H \text{ in } G\}$. This gives us a homomorphism $\pi_H : G \rightarrow S_G$. Now, by the previous proposition, $\ker \pi_H \trianglelefteq H$, so $\ker \pi_H \subseteq 1_G$. This means, $\ker \pi_H = 1_G$. Therefore, $G \cong \text{im } \pi_H \leq S_G$. $\square$

1.15. CLASSIFICATION OF FINITELY GENERATED ABELIAN GROUPS

We first recall the definition of a direct product of groups:

Definition 1.94 (Direct Product of Groups). *The direct product $G_1 \times G_2 \times \cdots \times G_n$ of the groups $G_1, \dots, G_n$ is the set*

$$\{(g_1, \dots, g_n) \mid g_i \in G_i\}$$

with the component-wise operation:

$$(g_1, \dots, g_n) \cdot (g'_1, \dots, g'_n) = (g_1g'_1, \dots, g_ng'_n).$$

Note that one can take the direct product of infinitely many groups as well.

Example 1.95. Let $G_i = \mathbb{R}$. Then, $\mathbb{R}^\times = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$.

Theorem 1.96. *If $G_1, G_2, \dots, G_n$ are groups, then*

$$|G_1 \times \cdots \times G_n| = |G_1||G_2| \cdots |G_n|.$$

In particular, the direct product $G_1 \times \cdots \times G_n$ is also a group.

Proposition 1.97. *Let $G_1, G_2, \dots, G_n$ be groups and let $G = G_1 \times \cdots \times G_n$.*

1. For each i ,

$$G_i \cong \{(1, \dots, 1, g_i, 1, \dots, 1) \mid g_i \in G_i\}.$$

Identifying the right hand side as G_i , we can say that

$$G_i \trianglelefteq G$$

and

$$G/G_i \cong G_1 \times \dots \times G_{i-1} \times G_{i+1} \times \dots \times G_n.$$

2. For each fixed i , define

$$\pi_i : G \rightarrow G_i$$

by

$$\pi_i(g_1, \dots, g_n) = g_i.$$

Then, π_i is a surjective homomorphism and

$$\ker \pi_i = \{(g_1, \dots, g_{i-1}, 1, g_{i+1}, \dots, g_n) \mid g_j \in G_j \forall j \neq i\}.$$

In particular,

$$\ker \pi_i \cong G_1 \times \dots \times G_{i-1} \times G_{i+1} \times \dots \times G_n.$$

3. Under the identifications in part (1), for all $x \in G_i$ and for all $y \in G_j$ with $j \neq i$, $xy = yx$. Here we are viewing G_i and G_j as normal subgroups of G .

Definition 1.98 (Finitely Generated Groups). A group G is called *finitely generated* (f.g.) if there is a finite subset $A \subseteq G$ such that $G = \langle A \rangle$.

Definition 1.99 (Free Abelian Group). For each $r \in \mathbb{Z}$ with $r \geq 0$, we let $\mathbb{Z}^r = \mathbb{Z} \times \dots \times \mathbb{Z}$ (where we define $\mathbb{Z}^0 = 1$). Then, $\mathbb{Z}^r$ is called the *free abelian group of rank r* .

Theorem 1.100 (Fundamental Theorem of Finitely Generated Abelian Groups). Let G be a finitely generated abelian group. Then,

$$G \cong \mathbb{Z}^r \times Z_{n_1} \times Z_{n_2} \times \dots \times Z_{n_s}$$

where $r, n_1, \dots, n_s \in \mathbb{Z}$ such that:

1. $r \geq 0$ and $n_j \geq 2$ for all j
2. $n_{i+1} \mid n_i$ for all $1 \leq i \leq s-1$.

In particular, the expression is unique in the following sense: if $G \cong \mathbb{Z}^q \times Z_{u_1} \times \dots \times Z_{u_t}$ such that the integers satisfy the conditions above, then $r = q$, $s = t$, and $n_i = u_i$.

Definition 1.101 (Free Rank, Invariant Factors). The integer r in Theorem 1.100 is called the *free rank* of the group G . The integers $n_1, \dots, n_s$ are called the *invariant factors* of G .

Observation. Recall that the cardinality of the direct product of a set of groups can be written as $|G_1 \times \dots \times G_n| = |G_1| \cdot |G_2| \cdot \dots \cdot |G_n|$. In light of this, we can see that in Theorem 1.100,

G is finite if and only if its free rank is 0, i.e. $r = 0$.

Observation. Suppose G is a finite abelian group. Then, we can see that, with the decomposition as in Theorem 1.100,:

$$|G| = \prod_{i=1}^s n_i.$$

Therefore, to classify all finite abelian groups of order n , we must find all sequences $n_1, \dots, n_s$ of integers such that:

1. $n_j \geq 2$ for all j
2. $n_{i+1} \mid n_i$ for all $i \in \{1, \dots, s-1\}$
3. $n_1 \cdots n_s = n$.

In particular, we also see that

- n_1 is the largest invariant factor
- each n_i must divide n
- if p is a prime factor of n , then it must also divide n_i for some $i \in \{1, \dots, s\}$
- if p divides n_j , then it must also divide all n_i for $i \leq j$
- **Every prime divisor of n must also divide n_1 .**
- In particular, if n is a product of **distinct primes**, then $n = n_1$.

Corollary 1.102. If G is a finite abelian group of order n such that n is the product of distinct primes, then G is isomorphic to Z_n . In other words, the only abelian group of order n is, up to isomorphism, Z_n .

Theorem 1.103 (Elementary Divisor Decomposition of f.g. abelian groups). Let G be a finitely generated abelian group of order $n > 1$ such that the following is the prime factorization of n :

$$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}.$$

Then,

1. $G \cong A_1 \times A_2 \times \cdots \times A_k$ where $|A_i| = p_i^{\alpha_i}$.
2. for each $A \in \{A_1, \dots, A_k\}$ such that $|A| = p^\alpha$, we can write:

$$A \cong Z_{p^{\beta_1}} \times Z_{p^{\beta_2}} \times \cdots \times Z_{p^{\beta_k}}$$

where $\beta_1 + \cdots + \beta_k = \alpha$ and $\beta_1 \geq \cdots \geq \beta_k \geq 1$.

3. This decomposition is unique in the sense that if $G \cong B_1 \times \cdots \times B_s$ with $|B_i| = p_i^{\alpha_i}$ for each i , then $B_i \cong A_i$ and $k = s$.

2. RINGS

2.1. DEFINITIONS AND IMMEDIATE PROPERTIES

Definition 2.1 (Rings). A ring is a set R with two operations, $+$ and $\times$, such that the following are satisfied:

1. $(R, +)$ is an abelian group,
2. $\times$ is an associative binary operation, i.e. $x \times (y \times z) = (x \times y) \times z$, for all $x, y, z \in R$
3. the distributive laws hold in R with respect to the two operations, i.e. $(x + y) \times z = (x \times z) + (y \times z)$ and $x \times (y + z) = (x \times y) + (x \times z)$.

Note that a ring is guaranteed to be an abelian group with respect to only the operation of addition, not multiplication. This means that there need not be a multiplicative identity and each $x \in R$ need not have an inverse with respect to multiplication.

Definition 2.2 (Commutative Rings, Unital rings). A ring such that multiplication is commutative is said to be a commutative ring. A ring with a multiplicative identity, $1_R \in R$ such that $1_R \times x = x \times 1_R = x$ for all $x \in R$, is said to be a unital ring.

Observation. Note that $(R, +)$ is forced to be abelian as a group by the distributive laws. Let R be a unital ring. Consider computing $(1 + 1)(a + b)$ by distributing over the first versus the second term. Then, one can arrive at the conclusion that $a + b = b + a$.

Definition 2.3 (Division Ring). A unital ring R with $1_R \neq 0$ such that every $x \in R$ has a multiplicative inverse (i.e. there exists $x^{-1} \in R$ such that $xx^{-1} = x^{-1}x = 1_R$) is called a division ring.

Definition 2.4 (Fields). A commutative division ring is called a field.

Example 2.5. The following are examples of rings:

1. The **trivial ring** R is any abelian group $(R, +)$ along with multiplication defined as $x \times y = 0$ for all $x, y \in R$. Note that the trivial ring does not have an identity.
2. The **zero ring** is $R = \{0\}$ with the trivial multiplication. **The zero ring is the only ring where the multiplicative identity is the same as the additive identity**, i.e. $1 = 0$.
3. $\mathbb{Z}$ is a commutative, unital ring with addition and multiplication defined on it as usual.
4. $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are commutative, unital rings. In particular, since every $x \in R$ has an inverse, these are fields.
5. $\mathbb{Z}/n\mathbb{Z}$ is a commutative, unital ring. Note that the multiplicative identity is simply $\bar{1}$

Example 2.6 (Real Hamilton Quaternions). Define the following set:

$$\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\}.$$

Define addition in a componentwise manner:

$$(a + bi + cj + dk) + (a' + b'i + c'j + d'k) = (a + a') + (b + b')i + (c + c')j + (d + d')k.$$

Define multiplication by expanding using the distributive laws and then simplifying using

$$i^2 = j^2 = k^2 = -1, \quad ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j.$$

Then, $\mathbb{H}$ is a non-commutative, unital ring where $1 = 1 + 0i + 0j + 0k$.

Example 2.7. The ring of even integers, $2\mathbb{Z}$, is a commutative ring without identity.

Proposition 2.8. Let R be a ring. Then,

1. $0r = r0 = 0$, for all $r \in R$
2. $(-a)b = -(ab) = a(-b)$ for all $a, b \in R$
3. $(-a)(-b) = ab$ for all $a, b \in R$
4. if R is a unital ring, then its identity is unique and $-a = (-1) \cdot a$ for all $a \in R$.

Definition 2.9 (Zero Divisors). Let R be a ring. Then, $x \in R$ is a zero divisor if $x \neq 0$ and there exists $y \neq 0$ such that either $xy = 0$ or $yx = 0$.

Examples 2.10. Here are some examples of zero divisors:

1. In $\mathbb{Z}/6\mathbb{Z}$, 3 is a zero divisor, since $3 \cdot 2 = 6 \equiv 0$.
2. In the ring $\{(a, b) \in \mathbb{Z}^2 \mid a \equiv b \pmod{3}\}$, $(3, 0)$ is a zero divisor, since $(3, 0) \cdot (0, 3) = (0, 0)$.

Definition 2.11 (Unit). Let R be a unital ring with $1 \neq 0$. An element $u \in R$ is called a unit if there exists $v \in R$ such that $uv = vu = 1$. The set of units of the ring R is denoted by $R^\times$.

Note that $(R^\times, \times)$ is a group under multiplication. This is called the **group of units of R** . Note that a field is a commutative, unital ring with $1 \neq 0$ where every non-zero element is a unit.

Lemma 2.12. A zero divisor can not be a unit.

Proof. Suppose $x \in R$ is a zero divisor in a unital ring with $1 \neq 0$. Then, $x \neq 0$ and there exists $y \neq 0$ such that $yx = 0$. Now, suppose x is a unit, i.e. there exists $z \in R$ such that $xz = zx = 1$. In particular, if $xz = 1$, then, $y = y \cdot 1 = y \cdot xz = (yx)z = 0 \cdot z = 0$ which is a contradiction. $\square$

Theorem 2.13. Let $n \geq 2$. Then, in the ring $\mathbb{Z}/n\mathbb{Z}$, the set of units are:

$$\mathbb{Z}/n\mathbb{Z}^\times = \{u \in \mathbb{Z}/n\mathbb{Z} \mid \text{g.c.d}(u, n) = 1\}.$$

On the other hand, the elements $v \in \mathbb{Z}/n\mathbb{Z}$ such that $\text{g.c.d}(v, n) > 1$ are zero divisors. As such, every element in $\mathbb{Z}/n\mathbb{Z}$ is either a zero divisor or a unit.

Proof. To prove the first statement, given $(u, n) = 1$, we know that there exists $x, y \in R$ such that $ux + ny = 1$. Therefore, $ux = 1 - ny$. Now, $\overline{ux} = \overline{1 - ny} = \overline{1}$.

To prove the second statement: let $d = (v, n) > 1$ and let $b = n/d \in (0, n)$, so $\bar{b} \neq \bar{0}$. Now, vb is divisible by n , so $\overline{vb} = 0$. $\square$

Theorem 2.14. $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is a prime.

Definition 2.15 (Integral Domains). A commutative, unital ring with $1 \neq 0$ such that it contains no zero divisors is called an integral domain. We will use the term integral domain and domain interchangeably.

Proposition 2.16. Let a, b, c be elements of a ring such that a is not a zero divisor. If $ab = ac$, then either $a = 0$ or $b = c$. In particular, integral domains have the cancellation property.

Corollary 2.17. Any finite integral domain is a field.

Proof. Let R be a finite integral domain. Let $a \neq 0$ be an element of this ring. Now consider the map $\varphi : R \rightarrow R$ by $x \mapsto ax$. Since R is finite and this map is injective (because of R being an integral domain), this map is surjective. Therefore, there exists some $b \in R$ such that $ab = 1$. Since a was arbitrarily chosen, R is a field. $\square$

Proposition 2.18 (Wedderburn). A finite division ring is necessarily commutative. Therefore, a finite division ring is a field.

Definition 2.19 (Subring). A subring of a ring R is a subgroup of R such that it is closed under multiplication.

Examples 2.20. The following are examples of subrings:

1. The set of even integers, $2\mathbb{Z}$, is a subring of $\mathbb{Z}$ which is a subring of $\mathbb{Q}$ which is a subring of $\mathbb{R}$. Any $n\mathbb{Z}$ is a subring of $\mathbb{Z}$ for any $n \in \mathbb{Z}$.

2.2. EXAMPLES OF RINGS

We start with an introduction to polynomial rings. Let R be a commutative, unital ring. Then, a polynomial is a formal sum:

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \quad n \geq 0, a_i \in R.$$

Then, the polynomial ring, $R[x]$, is the set of all such polynomials under the usual definition of addition and multiplication of polynomials.

We have the following vocabulary for describing polynomials:

1. Leading Term: if $a_n \neq 0$, then $a_n x^n$ is called the leading term.
2. Leading Coefficient: if $a_n \neq 0$, then a_n is called the leading coefficient.
3. Degree: the degree of the polynomial is n if its leading coefficient is a_n .
4. Monic Polynomial: a polynomial such that the leading coefficient is 1.

Observation. Note that R is a subring of $R[x]$. Also note that $R[x]$ is a commutative, unital ring.

Proposition 2.21. Let R be an integral domain (i.e. commutative, unital ring without zero divisors). Let $p(x), q(x) \in R[x]$ such that $p(x) \neq 0, q(x) \neq 0$. Then,

1. $\text{degree } p(x)q(x) = \text{degree } p(x) + \text{degree } q(x)$,
2. $R[x]^\times = R^\times$,
3. $R[x]$ is an integral domain.

Observation. If $R[x]$ is an integral domain, then R must be an integral domain, since R is a subring of $R[x]$.

Proposition 2.22. Let R be a commutative, unital ring. Then, $p(x) \in R[x]$ is a zero divisor if and only if there exists $c \in R, c \neq 0$, such that $cp(x) = 0$.

Proof. Suppose $p(x)$ is a zero divisor of degree n :

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0.$$

Now, suppose $g(x) \neq 0$ be the polynomial of minimal, non-zero degree such that $g(x)p(x) = 0$. Let

$$g(x) = b_m x^m + \cdots + b_1 x + b_0.$$

After taking the product, we see that $b_m a_n = 0$. As such, we see that $g(x)a_n$ has degree less than $m = \text{degree } g(x)$. Furthermore, $g(x)a_n \cdot p(x) = a_n g(x)p(x) = 0$. But this implies $a_n g(x) = 0$.

Since $a_n g(x) = 0$, this also means that $a_n b_{m-1} = 0$, so $a_n x^n b_{m-1} x^{m-1} = 0$. This means that $g(x)p(x) = 0$ implies $a_{n-1} b_m = 0$. This, similarly, implies that $a_{n-1} g(x) = 0$. Continuing this way, we see that $a_i b_m = 0$ for all $i \in \{1, \dots, n\}$. As such, $b_m p(x) = 0$. $\square$

Proposition 2.23. Define $R[[x]]$ to be the ring of formal infinite sums in the indeterminate x with coefficients from R :

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots.$$

Note that

$$\sum_{n=0}^{\infty} a_n x^n \times \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) x^n.$$

Then, $R[[x]]$ is a commutative, unital ring. In particular, $1-x$ is a unit in $R[[x]]$ with inverse $1+x+x^2+\cdots$. Furthermore, $\sum_{n=0}^{\infty} a_n x^n$ is a unit in $R[[x]]$ if and only if a_0 is a unit in R . If R is an integral domain then $R[[x]]$ is also an integral domain.

Definition 2.24 (Nilpotents). Let R be a ring. Then, $a \in R$ is called a nilpotent if there exists a positive integer n such that $a^n = 0$.

Definition 2.25 (Reduced Rings). A ring R is called reduced if the only nilpotent element in R is 0 itself.

Example 2.26. $\mathbb{Z}/8\mathbb{Z}$ is not a reduced ring as $2^3 = 8$.

Proposition 2.27. If R and S are reduced rings, then $R \times S$ is a reduced ring.

2.3. RING HOMOMORPHISMS AND QUOTIENT RINGS

Definition 2.28 (Ring homomorphisms, Kernels, and Isomorphisms). Let R and S be rings. Then, a ring homomorphism from R to S , $\varphi \in \text{Hom}(R, S)$, is a map $\varphi : R \rightarrow S$ such that

$$\varphi(x + y) = \varphi(x) + \varphi(y)$$

and

$$\varphi(xy) = \varphi(x)\varphi(y).$$

Although [1] do not require that ring homomorphisms send 1_R to 1_S , we will adopt that as a part of the definition. We will use the term ring homomorphisms and ring morphisms interchangeably. The kernel of the homomorphism is

$$\ker \varphi = \{x \in R \mid \varphi(x) = 0\}.$$

If φ is a bijection, we call it an isomorphism.

Examples 2.29. The following are examples and non-examples of ring homomorphisms.

1. $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ which sends evens to 0 and odds to 1.
2. For any $n \in \mathbb{Z}$, define the map

$$\varphi_n : \mathbb{Z} \rightarrow \mathbb{Z}$$

defined by

$$\varphi_n(x) = nx.$$

Then, φ_n is not a ring homomorphism, but is a group homomorphism.

3. The following is a ring homomorphism in $\text{Hom}(\mathbb{Q}[x], \mathbb{Q})$:

$$\varphi(p(x)) = p(0).$$

Proposition 2.30. The kernel and image are subrings. Let $\varphi : R \rightarrow S$ be a ring homomorphism. Then, $\text{im } \varphi \leq S$ and $\ker \varphi \leq R$.

Proposition 2.31. For any ring R , there exists a unique morphism $R \rightarrow \{0\}$. For any ring R , there exists a unique morphism $\mathbb{Z} \rightarrow R$. If ring morphisms are required to send 1 to 1, then there is only one ring homomorphism $\{0\} \rightarrow X$, and it is the morphism to itself (i.e. $X = \{0\}$).

Note that the last statement here comes from the fact that in all other rings $1 \neq 0$.

2.4. IDEALS, QUOTIENT RINGS, AND ISOMORPHISM THEOREMS

Definition 2.32 (Ideals). *Let R be a ring and let $I \subseteq R$ be a subset.*

1. *I is a left ideal if (i) $I \subseteq R$ is a subring, i.e. $I \leq R$ and (ii) $xI \subseteq I$ for all $x \in R$.*
2. *I is a right ideal if (i) $I \subseteq R$ is a subring, i.e. $I \leq R$, and (ii) $Ix \subseteq I$ for all $x \in R$.*
3. *I is an ideal if I is both a left ideal and right ideal. We denote this by $I \trianglelefteq R$.*

Proposition 2.33. *Let R be a ring and let $I \trianglelefteq R$. Then, the quotient group R/I is a ring under the operations;*

$$(x + I) + (y + I) = (x + y) + I$$

and

$$(x + I) \times (y + I) = (xy) + I.$$

Conversely, for any subgroup I of R such that these operations are well-defined, I is an ideal of R .

Definition 2.34 (Quotient Ring). *Let R be a ring and let $I \trianglelefteq R$ be an ideal of R . We call R/I a quotient ring.*

Note that $\bar{x} = \bar{y}$ in R/I if and only if $x + I = y + I$ if and only if $x - y \in I$. This follows from the results we showed for quotient groups.

Example 2.35. In any ring R , the subrings 0 and R are ideals. We call 0 the trivial ideal.

Proposition 2.36. *Let $\phi : R \rightarrow S$ be a ring morphism.*

1. *If $J \trianglelefteq S$, then $\phi^{-1}(J) \trianglelefteq R$.*
2. *if $I \trianglelefteq R$ and ϕ is surjective, then $\phi(I) \trianglelefteq S$.*
3. *In particular, $\phi^{-1}(0) =: \ker \phi \trianglelefteq R$.*

It is worth noting that, in (2), if ϕ is not surjective, then $\phi(I)$ need not be an ideal in S . For example, consider the ring morphism $\mathbb{Z} \rightarrow \mathbb{Z}[i]$, which maps $(2) \rightarrow 2\mathbb{Z}$. But $2\mathbb{Z}$ is not an ideal in $\mathbb{Z}[i]$ (since $i \cdot 2 \notin 2\mathbb{Z}$).

Proposition 2.37. *If $I \trianglelefteq R$ and $\phi : R \rightarrow S$ is a ring morphism with $\phi(I) = \{0\}$. Then, there exists a unique morphism $\bar{\phi} : R/I \rightarrow S$ such that $\bar{\phi} \circ \pi = \phi$ (where $\pi : R \rightarrow R/I$ is the natural morphism):*

$$\begin{array}{ccc} R & \xrightarrow{\pi} & R/I \\ & \searrow \phi & \downarrow \exists! \bar{\phi} \\ & & S \end{array}$$

In particular, for any ring morphism $\phi : R \rightarrow S$, there exists a unique morphism $\bar{\phi} : R/\ker \phi \rightarrow S$ such that $\bar{\phi} \circ \pi = \phi$.

Proposition 2.38 (Ring Isomorphism Theorems). *Let R be a ring.*

1. (1st Isomorphism Theorem) Let $\varphi \in \text{Hom}(R, S)$. Then, $\ker \varphi \trianglelefteq R$, $\text{im } \varphi \leq S$ (note: image is not necessarily an ideal, it is necessarily a subring), and $R/\ker \varphi \cong \text{im } \varphi$. Furthermore, if $I \trianglelefteq R$ is an ideal of R , then the map

$$R \rightarrow R/I$$

defined by

$$x \rightarrow x + I$$

is a surjective ring homomorphism with I its kernel. As such, every ideal is the kernel of some ring homomorphism and vice-versa.

2. (2nd Isomorphism Theorem) Let $A \leq R$ and $B \trianglelefteq R$. Then, $A+B \leq R$, $A \cap B \trianglelefteq R$, and $(A+B)/B \cong A/(A \cap B)$.
3. (3rd Isomorphism Theorem) Let $I \trianglelefteq R$ and let $J \trianglelefteq R$ such that $I \subseteq J$. Then, $J/I \trianglelefteq R/I$ and $(R/I)/(J/I) \cong R/J$.
4. (4th Isomorphism Theorem) Let $I \trianglelefteq R$. The correspondence

$$\{A \leq R \text{ s.t. } I \subseteq A\} \leftrightarrow \{A/I \leq R/I\}$$

is an inclusion-preserving bijection. In particular, $A \leq R$ with $I \subseteq R$ if and only if $A/I \leq R/I$.

Example 2.39. For any $n \in \mathbb{Z}$, $n\mathbb{Z} \trianglelefteq \mathbb{Z}$, and they are the only ideals of $\mathbb{Z}$.

Example 2.40. Let $R = \mathbb{Z}[x]$. Let I be the collection of polynomials whose individual terms are of degree greater than or equal to 2. Then, $I \trianglelefteq R$. Even though $R = \mathbb{Z}[x]$ has no zero divisors, R/I has:

$$\bar{x} \cdot \bar{x} = \bar{x}^2 = 0.$$

Example 2.41. For any m and n ,

$$m\mathbb{Z} + n\mathbb{Z} = d\mathbb{Z}, \text{ where } d = \gcd(m, n).$$

Definition 2.42 (Proper Ideal). An ideal $I \trianglelefteq R$ is called a *proper ideal* if $I \neq R$.

2.5. PROPERTIES OF IDEALS

Definition 2.43 (Ideal Generated by a Subset, Principal Ideal, Finitely Generated Ideal). Let R be a unital ring with $1 \neq 0$. Let $A \subseteq R$ be a subset.

1. We say (A) is the ideal generated by A , which is defined to be the smallest ideal of R containing A :

$$(A) = \bigcap_{I \trianglelefteq R, A \subseteq I} I.$$

In particular,

$$(A) = RAR = \{r_1 a_1 r'_1 + \cdots + r_n a_n r'_n \mid r_i, r'_i \in R, a_i \in A, n \in \mathbb{Z}_{>0}\}.$$

If R is also commutative, then

$$(A) = RAR = RA = AR.$$

2. An ideal generated by a single element is called a *principal ideal*. We will use the notation $(a) = (A)$ when $A = \{a\}$.
3. An ideal generated by a finite set is called a *finitely generated ideal*. We will use the notation $(a_1, \dots, a_n) := (A)$ where $A = \{a_1, \dots, a_n\}$.

Proposition 2.44. let $I \trianglelefteq R$ be an ideal and let R be a unital ring with $1 \neq 0$.

1. $I = R$ is and only if I contains a unit.
2. If R is commutative, then R is a field if and only if its only ideals are 0 and R .
3. $R = \{0\}$ if and only if it has only one ideal.

Corollary 2.45. *If R is a field, then any non-zero ring homomorphism from R to another ring is an injection.*

Examples 2.46. *The following are examples of various special ideas.*

1. The trivial ideal is a principal ideal: $0 = (0)$.
2. The ideal R is a principal ideal: $R = (1)$.
3. In $\mathbb{Z}$, the ideal $n\mathbb{Z} = \mathbb{Z}n = (n) = (-n)$ for any $n \in \mathbb{Z}$. Therefore, **every ideal of $\mathbb{Z}$ is a principal ideal**. Furthermore, $n\mathbb{Z} \subset m\mathbb{Z}$ if and only if m divides n .
4. In the ring $\mathbb{Z}[x]$, the ideal $(2, x)$ is not a principal ideal.

Proposition 2.47. *The ideals of $R \times S$ are of the form $I \times J$ where $I \trianglelefteq R$ and $J \trianglelefteq S$.*

Proof. For any ideal $K \trianglelefteq R \times S$, suppose $(a, b) \in K$. Now,

$$(1, 0) \cdot (a, b) = (a, 0) \in K$$

and

$$(0, 1) \cdot (a, b) = (0, b) \in K$$

since K is an ideal. Now, define $I = \{a \in R \mid (a, 0) \in K\}$ and $J = \{b \in S \mid (0, b) \in K\}$. The properties of ideals forces I to be an ideal in R and J to be an ideal in S . On the other hand, $K = I \times J$. $\square$

2.6. MAXIMAL IDEALS

Definition 2.48 (Maximal Ideals). *An ideal $M \trianglelefteq S$ is called a maximal ideal if $M \neq S$ and the only ideals containing M are M itself and S .*

Proposition 2.49. *In a unital ring, every proper ideal is contained in a maximal ideal.*

Proof. The proof uses Zorn's lemma. Let I be a proper ideal and define

$$\mathcal{S} = \{J \trianglelefteq R \mid I \subseteq J, J \neq R\}.$$

Now,

$$\mathcal{S} \neq \emptyset$$

since $I \in \mathcal{S}$. Also $\mathcal{S}$ is partially ordered by inclusion. Now, define $\mathcal{C}$ to be a chain inside $\mathcal{S}$ and define $J = \cup_{A \in \mathcal{C}} A$. Then, it is easy to check that J is itself an ideal. Furthermore, J is a proper ideal: if J weren't a proper ideal, then $1 \in J$ which would imply that $1 \in A$ for all $A \in \mathcal{C}$, which is a contradiction since $\mathcal{C}$ is in $\mathcal{S}$, which consists of only proper ideals. As such, every chain $\mathcal{C}$ has an upper bound which belongs in $\mathcal{S}$. By Zorn's lemma, $\mathcal{S}$ has a maximal element which is our maximal ideal. $\square$

Proposition 2.50. *Let R be a commutative, unital ring with $1 \neq 0$. Then, the ideal M is a maximal ideal if and only if R/M is a field.*

Proof. $M \trianglelefteq R$ is a maximal ideal if and only if there are no ideals I such that $M \subset I \subset R$. By the correspondence theorem, the ideals of R containing M correspond to the ideals of R/M . Therefore, $M \trianglelefteq R$ is a maximal ideal if and only if the only ideals of R/M are 0 and R/M , which is true if and only if R/M is a field. $\square$

Examples 2.51. The following are examples (and non-examples) of maximal ideals.

1. Let $n \in \mathbb{Z}_{\geq 0}$. Then, $n\mathbb{Z} \trianglelefteq \mathbb{Z}$ is a maximal ideal if and only if n is a prime.
2. $(2, x)$ is a maximal ideal in $\mathbb{Z}[x]$ as $\mathbb{Z}[x]/(2, x) \cong \mathbb{Z}/2\mathbb{Z}$.

Proof. Inside $\mathbb{Z}[x]/(2, x)$, 2 is 0 since $2 + (2, x) \in (2, x)$. Furthermore, x is 0 since $x + (2, x) \in (2, x)$. Therefore, the only elements in $\mathbb{Z}[x]/(2, x)$ are 0 and 1. $\square$

3. (x) is not a maximal ideal in $\mathbb{Z}[x]$ as $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, which is not a field.

2.7. PRIME IDEALS

Definition 2.52 (Prime Ideals). Let R be a commutative ring. Then, the ideal $P \trianglelefteq R$ is called a prime ideal if $P \neq R$ and $xy \in P$ implies that either $x \in P$ or $y \in P$.

Observation. Consider the ring $\mathbb{Z}$. Then, for $n \neq 1$, the ideal $n\mathbb{Z}$ is a prime ideal if n is a prime: $n\mathbb{Z}$ is prime is $xy \in n\mathbb{Z}$ implies either $x \in n\mathbb{Z}$ or $y \in n\mathbb{Z}$. So, $n|xy$ implies that $n|x$ or $n|y$. This condition is equivalent to the condition that n is a prime. Therefore, **the prime ideals in $\mathbb{Z}$ are the trivial ideal and $p\mathbb{Z}$ for primes p .**

Proposition 2.53. Suppose R is a commutative ring. Then, the ideal P is a prime ideal in R if and only if R/P is an integral domain.

Proof. P is a prime ideal if and only if $xy \in P$ implies $x \in P$ or $y \in P$ and $R \neq P$. This is equivalent to the condition that, in R/P , $\bar{x}\bar{y} = 0$ implies $\bar{x} = 0$ or $\bar{y} = 0$ and $\bar{R} \neq 0$. As such, there are no zero divisors, which, by definition, means R/P is an integral domain. $\square$

Corollary 2.54. Suppose R is a commutative ring. Then, any maximal ideal of R is a prime ideal.

Examples 2.55. The following are examples of prime ideals.

1. The principal ideals generated by primes in $\mathbb{Z}$ are both prime and maximal ideals. The zero ideal in $\mathbb{Z}$ is prime but not maximal.
2. $(x) \trianglelefteq_{\text{prime}} \mathbb{Z}[x]$ as $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$. However, (x) is not a maximal ideal.
3. $(0) \trianglelefteq_{\text{prime}} \mathbb{Z}[x]$ but (0) is not a maximal ideal.

Definition 2.56 (Spectrum of a ring). Define the spectrum of a ring to be the set of all prime ideals of the ring, i.e.

$$\text{Spec}(R) = \{P \mid P \trianglelefteq_{\text{prime}} R\}.$$

Theorem 2.57. If $\phi : R \rightarrow S$ is a ring homomorphism. If $J \trianglelefteq S$ is a prime ideal, then $\phi^{-1}(J)$ is a prime ideal in R . In other words,

$$\phi^{-1} : \text{Spec}(S) \rightarrow \text{Spec}(R).$$

If ϕ is an isomorphism, then there is a bijection between the spectrums of R and S are isomorphic.

Proof. Suppose J is a prime ideal of S . Then, for any $s_1 s_2 \in J$, either $s_1 \in J$ or $s_2 \in J$. Now, consider $r_1 \cdot r_2 \in \phi^{-1}(J)$. Then, if neither r_1 nor r_2 are in $\phi^{-1}(J)$, then we get $\phi(r_1 \cdot r_2) = \phi(r_1)\phi(r_2) \in J$ where neither $\phi(r_1)$ nor $\phi(r_2)$ is in J , which would be a contradiction. So either r_1 or r_2 is in $\phi^{-1}(J)$. $\square$

Observation. This statement would not hold true for maximal ideals. Consider the homomorphism $\mathbb{Z} \rightarrow \mathbb{Q}$. Since $\mathbb{Q}$ is a field, 0 is a maximal ideal, but its preimage is $(0) \subsetneq (2) \subsetneq \mathbb{Z}$.

Theorem 2.58. The prime ideals of $R \times S$ are the ideals $I \times S$ for $I \trianglelefteq_{\text{prime}} R$ and $R \times J$ for $J \trianglelefteq_{\text{prime}} S$.

Proof. Let P be a prime ideal of $R \times S$. Now, $(1, 0) \cdot (0, 1) = (0, 0) \in P$ so, either $(1, 0)$ or $(0, 1)$ is in P .
Case 1: $(1, 0) \in P$. Then, $(r, 0) = (r, 0) \cdot (1, 0) \in P$, so

$$R \times \{0\} \subseteq P.$$

Now, let $J = \{s \in S \mid (0, s) \in P\}$, which is an ideal of S . Then, $P = R \times J$ since for any $(r, s) \in P$, we have that $(0, s) = (r, s) - (r, 0) \in P$, so $s \in J$, which implies $(r, s) \in R \times J$.

Case 2: $(0, 1) \in P$. This case, similarly, implies that $P = I \times S$ where $I = \{r \in R \mid (0, r) \in P\}$. $\square$

Theorem 2.59. There exists the following bijection:

$$\{\text{prime ideals of } R/I\} \leftrightarrow \{\text{prime ideals of } R \text{ containing } I\}.$$

Proof. Define $\pi : R \rightarrow R/I$ by $r \mapsto r + I$. Now, if $\bar{P} \trianglelefteq_{\text{prime}} R/I$, then $\pi^{-1}(\bar{P})$ is a prime ideal in R . Furthermore, $\ker \pi = I \in \pi^{-1}(\bar{0})$, since it maps to 0 which is in $\bar{P}$. So, we have mapped a prime ideal $\bar{P}$ back to a prime ideal in R containing I . Conversely, suppose P is prime in R and $I \subseteq P$. Then, $\pi(P) = P/I$. We claim that P/I is prime: suppose $(a + I) \cdot (b + I) \in P/I \implies (ab + I) \in P/I$. Now, since $ab \in P$, either $a \in I$ or $b \in P$, so either $a + I \in P/I$ or $b + I \in P/I$. $\square$

2.7.1 Set of Nilpotents and Intersection of Prime Ideals

We will show that the set of nilpotent elements of a ring equals to the intersection of all ideals in the spectrum of a ring. We will require the following lemma:

Lemma 2.60. Suppose that I is an ideal of a ring R and that $S \subset R - I$ is a subset. Then, R contains a maximal ideal J containing I but not containing any element of S .

Proof. Consider the set $X = \{K \trianglelefteq R \mid I \subseteq K, K \cap S = \emptyset\}$. This set is partially ordered and non-empty since $I \in X$. Now, consider a chain of ideals $\mathcal{C} \subset X$. Then, consider the element

$$J = \cup_{K \in \mathcal{C}} K.$$

J is clearly an ideal (the interesting part is to show that if, $a, b \in J$, then $a + b \in J$; this follows because $a \in K_n$ and $b \in K_m$, then, $a + b \in K_n$ if $K_m \subseteq K_n$, otherwise $a + b \in K_m$). Next, we can see that $I \subset J$ since $I \subseteq K$ for each K in the chain. Furthermore, none of the ideals in the chain intersect with S , so S does not intersect with J . Therefore, $J \in X$. So, we have an upper bound for any chain in X that is inside X itself. So, by Zorn's lemma, X has a maximal element J that contains I and does not intersect with S . $\square$

We will also need the following lemma:

Lemma 2.61. Suppose now that $x \in R$ is not nilpotent and $I = (0)$ and $S = \{1, x, x^2, \dots\}$. Let J be a maximal ideal not intersecting with S , which exists by the previous lemma. Show that J is a prime ideal.

Proof. Let J be our maximal ideal not intersecting with I . Now, suppose $ab \in J$. We show that either a or b is in J . Suppose, for contradiction, neither a nor b is in J . Then, $J \subsetneq J + (a)$. Since $J + (a)$ is a larger ideal containing J , it must intersect with S . Then, there exists some n such that $x^n \in S$ such that $x^n \in J + (a)$, i.e. $x^n = j + ra$ for some $j \in J, r \in R$ and $n \in \mathbb{Z}_{\geq 0}$. Similarly, there exists some $m \in \mathbb{Z}_{\geq 0}$ such that $x^m = j' + sb$ for some $j' \in J$ and $s \in R$. Now, one can show that $x^{n+m} \in J$, which would show that $J \cap S \neq \emptyset$, which is a contradiction, so J is a prime ideal. $\square$

Finally, we come to the main result of this section:

Proposition 2.62. *The intersection of all prime ideals of R equals the set of nilpotent elements of R . In other words, $\cap_{P \in \text{Spec}(R)} P = \{r \in R \mid r \text{ nilpotent}\}$.*

Proof. Consider a nilpotent r . Then, r is nilpotent shows that $r^n = 0$ for some $n \in \mathbb{Z}_{\geq 0}$. But 0 is an element in every prime ideal, so $0 \in \cap_{P \in \text{Spec}(R)} P$, so $r^n \in \cap_{P \in \text{Spec}(R)} P$. But then, $r^n \in P$ for each prime ideal P , which shows that $r \in P$ for each prime ideal P . So, $r \in \cap_{P \in \text{Spec}(R)} P$. On the other hand, suppose $r \in \cap_{P \in \text{Spec}(R)} P$. We show that r is nilpotent. We show this through a contrapositive: we will prove that if r is not nilpotent, then r is not in $\cap_{P \in \text{Spec}(R)} P$. Given r is not nilpotent, we have that $J \cap S = \emptyset$ where $S = \{1, r, r^2, \dots\}$. So J does not contain r , and so, we have a prime ideal that does not contain r . Therefore, $r \notin \cap_{P \in \text{Spec}(R)} P$. $\square$

2.8. LOCALISATION AND RING OF FRACTIONS

In this section, we will show that a commutative ring R is a subring of a larger ring Q in which every element can become a unit.

Definition 2.63 (Multiplicative Subset). *Let R be a unital ring. Then, D is a multiplicative subset of the ring R if:*

1. $1 \in D$, and
2. D is closed under multiplication.

Definition 2.64 (Ring of Fractions). *Let R be a unital ring and let D be a multiplicative subset. Then, define*

$$D^{-1}R := \left\{ \frac{r}{d} \mid r \in R, d \in D \right\},$$

and define the equivalence relation:

$$\frac{r_1}{d_1} = \frac{r_2}{d_2} \text{ if and only if } \exists f \in D, \text{ s.t. } f(r_1 d_2 - r_2 d_1) = 0.$$

Then, $D^{-1}R$ is called the ring of fractions or the localisation of R by D .

Proposition 2.65. *If $0 \in D$, then $D^{-1}R \cong 0$, i.e. the localisation is the zero ring.*

Proposition 2.66. *We have the canonical ring morphism $R \rightarrow D^{-1}R$ which sends $r \rightarrow \frac{r}{1}$. Furthermore, this homomorphism is injective if and only if D contains no zero divisors.*

Examples 2.67. *Here are some examples of ring morphisms:*

1. *If $S = \{1, f, f^2, f^3, \dots\}$, then*

$$A_f := S^{-1}A = \{1, f, f^2, \dots\}^{-1}A.$$

2. *If P is a prime ideal of ring A , then, define $S := A - P$. We have the localisation*

$$A_P = S^{-1}A = (A - P)^{-1}A.$$

3. *If A is an integral domain, then with $S = A - \{0\}$, we have the **field of fractions**:*

$$K(A) = S^{-1}A = (A - \{0\})^{-1}A.$$

As the name suggests, $K(A)$ is a field. In particular, $\mathbb{Q} = K(\mathbb{Z})$.

This last example, that of field of fractions, can be noted especially through the following proposition:

Theorem 2.68. *The ring R is an integral domain if and only if R embeds as a subring of a field. Note that this result does not hold if R is not commutative.*

Proposition 2.69. *Let R be a commutative ring with $1 \neq 0$. Then,*

1. $R[x]/(x - a) \cong R$ for any $a \in R$.

2. $R[x]/(fx-1) \cong R_f := \{1, f, f^2, \dots\}^{-1}R$ where $f \in R$ is fixed. If f is a unit, then, $R[x]/(fx-1) \cong R$.
3. Let A be any commutative, unital ring and let $a \in A$ be fixed. Then, $A[y]/(y-a) \cong A$.
4. $R[x, y]/(y-g(x)) \cong R[x]$.
5. $R[x_1, \dots, x_n]/(x_1-a_1, \dots, x_n-a_n) \cong R$ for fixed $a_1, \dots, a_n \in R$.
6. Let $I \trianglelefteq R$. Then, $R[x]/IR[x] \cong (R/I)[x]$. More generally, $R[x_1, \dots, x_n]/IR[x_1, \dots, x_n] \cong (R/I)[x_1, \dots, x_n]$.
7. If a non-zero integer D is not a perfect square, then $\mathbb{Z}[\sqrt{D}] \cong \mathbb{Z}[x]/(x^2-D)$.

Proposition 2.70. Let $\phi : R \rightarrow S$ be a ring homomorphism. Let $D \subset R$ be a multiplicative subset, and suppose $\phi(D) \subset S^\times$. Then, there exists a unique $\tilde{\phi} : D^{-1}R \rightarrow S$ such that $\phi = \tilde{\phi} \circ \iota$ (where $\iota : R \rightarrow D^{-1}R$ is the natural ring morphism):

$$\begin{array}{ccc} R & \xrightarrow{\iota} & D^{-1}R \\ & \searrow \phi & \downarrow \exists! \tilde{\phi} \\ & & S. \end{array}$$

As an “application”, we will prove the following using the universal property:

Proposition 2.71. If $D \subset R^\times$, then $D^{-1}R = R$.

Proof. Define S to be R in the universal property. Then, $\tilde{\phi} \circ \iota = \text{id}$, for some unique $\tilde{\phi}$, by the commutativity of the diagram. We can check that if we define $\tilde{\phi}(\frac{a}{d}) = ad^{-1}$, we satisfy the diagram. Also, notice that $\tilde{\phi}$ is surjective, since for any $r \in R$, $\tilde{\phi}(\frac{r}{1}) = r$. Now, on the other hand, if we set S to be $D^{-1}R$ in the universal property, then we have that:

$$\begin{array}{ccc} R & \xrightarrow{\iota} & D^{-1}R \\ & \searrow \iota & \downarrow \exists! \Psi \\ & & D^{-1}R. \end{array}$$

In particular, since $\tilde{\phi} \circ \iota = \text{id}_R$, we can set $\Psi = \iota \circ \tilde{\phi}$, because then $\iota \circ \tilde{\phi} \circ \iota = \iota \circ \text{id}_R = \iota$. By uniqueness of the universal property, $\iota \circ \tilde{\phi} = \text{id}_{D^{-1}R}$. This shows that $\tilde{\phi}$ is an isomorphism between $D^{-1}R$ and R . $\square$

2.8.1 Example Computations

We will go through a series of computations here that tie together various parts of the notes so far.

$$1. \mathbb{Z}[\sqrt{-3}]/(2) \cong \mathbb{Z}[x]/(2, x^2 + 3).$$

Proof. First, note that $\mathbb{Z}[\sqrt{-3}]/(2) \cong \mathbb{Z}/2\mathbb{Z}[\sqrt{-3}]$. This is simply because in $\mathbb{Z}[\sqrt{-3}]/(2)$, each element is of the form $a + b\sqrt{-3} + (2)$, and so $a, b \in \{0, 1\}$. Now, consider the morphism $\varphi : \mathbb{Z}[x] \rightarrow \mathbb{Z}/2\mathbb{Z}[\sqrt{-3}]$ using $\varphi(a) = \bar{a} \in \mathbb{Z}/2\mathbb{Z}$ and $\varphi(x) = \sqrt{-3}$. Then, this is surjective and its kernel is the ideal $(2, x^2 + 3) \subset \mathbb{Z}[x]$. $\square$

$$2. \mathbb{Z}[x]/(2, x^2 + 3) \cong \mathbb{F}_2[x]/(x^2 + 1).$$

Proof. First, we know from the definitions that $\mathbb{Z}[x]/(2, x^2 + 3) \cong \mathbb{F}_2[x]/(x^2 + 3)$. Now, consider the ideal $(x^2 + 3) \subset \mathbb{F}_2[x]$. But in $\mathbb{F}_2[x]$, $\bar{3} = \bar{1}$, so $(x^2 + 3) = (x^2 + 1)$. $\square$

$$3. \mathbb{F}_2[x]/(x^2 + 1) \cong \mathbb{F}_2[x]/((x + 1)^2).$$

Proof. Consider the ideal $((x + 1)^2) \subset \mathbb{F}_2[x]$. Now, in $\mathbb{F}_2[x]$, $(x + 1)^2 = x^2 + 2x + 1 \equiv x^2 + 1$, which implies the statement. $\square$

$$4. \mathbb{Z}[\sqrt{-3}]/(2) \text{ is not reduced, since it is isomorphic to } \mathbb{F}_2[x]/((x + 1)^2) \text{ in which } (x + 1) \neq 0 \text{ is nilpotent.}$$

$$5. \mathbb{Z}[\sqrt{-3}]/(2) \text{ is not an integral domain or a field.}$$

$$6. \text{The only maximal ideal of } \mathbb{Z}[\sqrt{-3}]/(2) \text{ is } \overline{(1 + \sqrt{-3})}.$$

Proof. We compute the maximal ideals of $\mathbb{F}_2[x]/((x + 1)^2)$. A maximal ideal, M , must contain $0 \equiv (x + 1)^2$. Since R is commutative and unital, the maximal ideal is also prime, and therefore, $\overline{x + 1} \in M$. Now, the ideal generated by this element must be in M , i.e. $\overline{(x + 1)} \subset M$. However, $\overline{(x + 1)}$ itself is a maximal ideal in $\mathbb{F}_2[x]/((x + 1)^2)$:

$$\begin{aligned} (\mathbb{F}_2[x]/((x + 1)^2)) / \overline{(x + 1)} &= (\mathbb{F}_2[x]/((x + 1)^2)) / ((x + 1)/((x + 1)^2)) \\ &= \mathbb{F}_2[x]/(x + 1) \quad (\text{by 3rd isomorphism theorem}) \\ &= \mathbb{F}_2, \end{aligned}$$

which is a field. So, we have that $\overline{(x + 1)}$ is a maximal ideal contained inside M , so M itself must be $\overline{(x + 1)}$.

Now, we aim to find the ideal in $\mathbb{Z}[\sqrt{-3}]/(2)$ that corresponds to $\overline{(x + 1)}$ in $\mathbb{F}_2[x]/((x + 1)^2)$. First, we find the corresponding ideal in $\mathbb{Z}[x]/(2, x^2 + 3)$. Recall the map:

$$\mathbb{Z}[x]/(2, x^2 + 3) \rightarrow \mathbb{F}_2[x]/((x + 1)^2)$$

via taking every integer modulo 2. Then, $f(x) + (2, x^2 + 3) \rightarrow \overline{f(x)} + ((x + 1)^2)$ (here, $\overline{f(x)}$ is simply $f(x)$ modulo 2). So if $\overline{f(x)} + ((x + 1)^2) \in \overline{(x + 1)} = (x + 1 + ((x + 1)^2))$, then $\overline{f(x)} \in (x + 1)$ (given $((x + 1)^2) \subset (x + 1)$). So, $\overline{f(x)} = \overline{q(x)} \cdot (x + 1) \in \mathbb{F}_2[x]$, which implies

$$f(x) - q(x)(x + 1) \in (2),$$

so

$$f(x) \in (2, x + 1).$$

Conversely, we claim that if $f(x) \in (2, x+1)$, then its image lies in $(\overline{x+1}) = (x+1 + ((x+1)^2))$. Indeed, if $f(x) = 2a(x) + (x+1)b(x)$, then it maps to (after taking modulo 2) $\overline{f(x)} = \overline{b(x)} \cdot (x+1)$, which proves the statement.

So, the corresponding maximal ideal in $\mathbb{Z}[x]/(2, x^2 + 3)$ is $(\overline{2, x+1})$. In $\mathbb{Z}[\sqrt{-3}]/(2)$, this ideal corresponds to $(1 + \sqrt{-3})$. $\square$

Problem 2.72. For each of the following rings, determine whether they are (a) reduced, (b) integral domains, and (c) fields:

$\mathbb{Z}[x]/(x^2-1)$; $\mathbb{Z}[x]/(x-1)^2$; $\mathbb{Z}[x]/(x^2+1)$; $\mathbb{Z}[\sqrt{-5}]/(2)$; $\mathbb{Z}[\sqrt{-5}]/(3)$; $\mathbb{Z}[\sqrt{-5}]/(1+\sqrt{-5})$; $\mathbb{Z}[\sqrt{-5}]/(11)$.

Answer. (1) $\mathbb{Z}[x]/(x^2 - 1) = \mathbb{Z}[x]/((x+1)(x-1))$. This ring is clearly not an integral domain since $x+1$ and $x-1$ are nonzero elements but their product is 0. Therefore, it is also not a field. Now, suppose $f(x) \neq 0$ in this ring and $f(x)^n \equiv 0$. Then, $f(x)^n \in ((x+1)(x-1))$. Now, $x+1$ is prime (since it is irreducible in the PID $\mathbb{Z}[x]$), and since $x+1$ divides $f(x)^n$, it must divide $f(x)$. Similarly, $x-1$ is also prime and therefore it must divide $f(x)$. Since $x+1$ is not associate to $x-1$, $f(x) = r(x)(x+1)(x-1)$ for some $r(x) \in \mathbb{Z}[x]$, so $f(x) \equiv 0$. Therefore, the only nilpotent element is 0 itself and so, the ring is reduced.

(2) $\mathbb{Z}[x]/(x-1)^2$ has a non-zero nilpotent element, namely $(x-1)$. So, this ring is not reduced. Therefore, it is also not an integral domain and, therefore, not a field.

(3) $\mathbb{Z}[x]/(x^2 + 1)$. Note that, in $\mathbb{Z}[x]$, $x^2 + 1$ is an irreducible polynomial since -1 is not a square. $\mathbb{Z}[x]$ is a PID, so $(x^2 + 1)$ is a prime ideal, and so $\mathbb{Z}[x]/(x^2 + 1)$ is an integral domain. However, it is not a maximal ideal since $(x^2 + 1) \subsetneq (2, x^2 + 1) \subsetneq \mathbb{Z}[x]$. So, the ring is not a field. Finally, we claim that the ring is reduced because we have shown that $x^2 + 1$ is irreducible: suppose $f(x)^n \equiv 0$. Then, $f(x) \in (x^2 + 1)$. But $x^2 + 1$ is prime, so $f(x) \equiv 0$.

(4) Consider the ring: $\mathbb{Z}[\sqrt{-5}]/(2)$. Consider $\mathbb{Z}[x]/(2, x^2+5) \cong \mathbb{Z}/2\mathbb{Z}[x]/(x^2+5) = \mathbb{F}_2[x]/(x^2+5) = \mathbb{F}_2[x]/(x^2+1) = \mathbb{F}_2[x]/(x+1)^2$, where the last equality follows from $(x+1)^2 = x^2 + 2x + 1 \equiv x^2 + 1$. In this ring, $(x+1)$ is nonzero but nilpotent, so the ring is not reduced. Therefore, it is also not an integral domain and not a field. Therefore, these statements hold for $\mathbb{Z}[\sqrt{-5}]/(2) \cong (\mathbb{Z}[x]/(x^2 + 5))/(2) \cong \mathbb{Z}[x]/(2, x^2 + 5)$, where the last isomorphism follows from the third isomorphism theorem: In $(\mathbb{Z}[x]/(x^2 + 5))$, the ideal $(\bar{2}) = \{2f(x) + (x^2 + 5) \mid f(x) \in \mathbb{Z}[x]\}$. As an ideal in $\mathbb{Z}[x]$, this is $(2, x^2 + 5)$. Since we are quotienting by $(x^2 + 5)$, this becomes

$$(\mathbb{Z}[x]/(x^2 + 5))/(2, x^2 + 5)/(x^2 + 5) \cong \mathbb{Z}[x]/(2, x^2 + 5).$$

(5) Consider the ring $\mathbb{Z}[\sqrt{-5}]/(3)$. Then, similar to the previous problem, $\mathbb{Z}[\sqrt{-5}]/(3) \cong (\mathbb{Z}[x]/(x^2 + 5))/(3) \cong \mathbb{Z}[x]/(3, x^2 + 5) \cong \mathbb{F}_3[x]/(x^2 + 5) = \mathbb{F}_3[x]/(x^2 + 2) = \mathbb{F}_3[x]/(x^2 - 1) = \mathbb{F}_3[x]/((x+1)(x-1))$. Now, $(x+1)$ and $(x-1)$ are nonzero but their product is 0, so we have zero divisors, implying that this is neither an integral domain nor a field. Suppose $f(x)^n \equiv 0$, i.e. $f(x)^n = g(x)(x+1)(x-1)$ for some $g(x) \in \mathbb{F}_3[x]$. Now, $(x+1)$ is irreducible and $\mathbb{F}_3[x]$ is a UFD, so it is also prime, so $(x+1)$ must divide $f(x)$. Similarly, $(x-1)$ must also divide $f(x)$. But these two factors are not associate, so $f(x)$ must be divisible by $(x+1)(x-1)$, therefore, $f(x) \equiv 0$ in our ring, meaning there are no non-zero nilpotent elements. As such, our ring is reduced.

(6) Consider the ring, $\mathbb{Z}[\sqrt{-5}]/(1+\sqrt{-5})$. Firstly, again, $\mathbb{Z}[\sqrt{-5}] \cong \mathbb{Z}[x]/(x^2+5)$. Then, $(\mathbb{Z}[x]/(x^2+5))/(1+\sqrt{-5}) \cong \mathbb{Z}[x]/(x^2+5, 1+\sqrt{-5}) = \mathbb{Z}[x]/(x^2+5, 1+x)$.

Now,

$$\begin{aligned} \mathbb{Z}[x]/(x^2+5, 1+x) &\cong (\mathbb{Z}[x]/(1+x))/(\overline{x^2+5}) && \text{(3rd isomorphism theorem)} \\ &\cong \mathbb{Z}/(\overline{x^2+5}) && \text{(using } \mathbb{Z}[x]/(1+x) \cong \mathbb{Z} \text{ via evaluating } x = -1) \\ &\cong \mathbb{Z}/(\overline{6}) \\ &= \mathbb{Z}/6\mathbb{Z}. \end{aligned}$$

Since 6 is not a prime, this is not a field and also not an integral domain, since $\bar{3} \cdot \bar{2} = 0$. This ring is reduced: $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ by Chinese Remainder Theorem. Both $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ are fields, so they are reduced. The product of reduced rings is reduced (from lecture 1), so our ring is reduced.

(7) Consider the ring $\mathbb{Z}[\sqrt{-5}]/(11)$. Again, we simplify the ring:

$$\begin{aligned}\mathbb{Z}[\sqrt{-5}]/(11) &\cong (\mathbb{Z}[x]/(x^2 + 5))/(\overline{11}) \\ &\cong \mathbb{Z}[x]/(11, x^2 + 5) \\ &\cong \mathbb{F}_{11}[x]/(x^2 + 5) \\ &\cong \mathbb{F}_{11}[x]/(x^2 - 6).\end{aligned}$$

Now, 6 is not a square modulo 11, so $x^2 - 6$ is irreducible and, therefore, prime (as $\mathbb{F}_{11}[x]$ is a PID). Then, this ring is reduced since $f(x)^n \in (x^2 - 6)$ implies $f(x) \in (x^2 - 6)$. Since this is a prime ideal, $\mathbb{F}_{11}[x]/(x^2 - 6)$ is an integral domain. In PIDs, prime ideals are maximal ideals, so this is also a field.

2.9. CHINESE REMAINDER THEOREM

In this section, we let R be a commutative, unital ring with $1 \neq 0$.

Recall the definition of a direct product of rings:

Definition 2.73 (Direct Product of Rings). *Let R and S be rings. Then, their direct product is:*

$$R \times S = \{(r, s) \mid r \in R, s \in S\}.$$

Definition 2.74 (Comaximal Ideals). *The ideals $A \trianglelefteq R$ and $B \trianglelefteq R$ are comaximal if $A + B = R$.*

Theorem 2.75 (Chinese Remainder Theorem). *Let R be a commutative, unital ring with $1 \neq 0$. Let $A_1, A_2, \dots, A_k$ be ideals in the ring R . Then, the map*

$$\varphi : R \rightarrow R/A_1 \times R/A_2 \times \dots \times R/A_k$$

is a ring homomorphism. In particular:

1. $\ker \varphi = A_1 \cap A_2 \cap \dots \cap A_k$ is the kernel.
2. If $\forall i, j \in [k]$ such that $i \neq j$, A_i and A_j are comaximal, then φ is surjective, and $A_1 \cap A_2 \cap \dots \cap A_k = A_1 A_2 \dots A_k$. So,

$$R/A_1 \dots A_k = R/(A_1 \cap A_2 \cap \dots \cap A_k) \cong R/A_1 \times R/A_2 \times \dots \times R/A_k.$$

Proof. We prove this via induction, starting with $k = 2$. The first part is obvious, via the map $\varphi(r) = (r + A, r + B)$. Now, suppose A and B are comaximal. Then, there exists $x \in A, y \in B$ such that $x + y = 1$. Now, $\varphi(x) = (0, 1)$ as $x \in A$ and $x = 1 - y \in 1 + B$. On the other hand, $\varphi(y) = (1, 0)$. This creates our *basis*: for any $(r_1 + A, r_2 + B)$, we have that:

$$\begin{aligned} \varphi(r_2 x + r_1 y) &= \varphi(r_2 x) + \varphi(r_1 y) \\ &= \varphi(r_2) \varphi(x) + \varphi(r_1) \varphi(y) \\ &= (r_2 + A, r_2 + B)(0, 1) + (r_1 + A, r_1 + B)(1, 0) \\ &= (r_1 + A, r_2 + B). \end{aligned}$$

So, φ is surjective. One can easily show that $AB = A \cap B$.

Now, for any general $k - 1$, suppose the statement is true. Then, for k : define $A = A_1$ and $B = A_2 \dots A_k$. Then, we claim that A and B are comaximal. To prove this, note that inductive hypothesis says that A_1 and A_i for $i \neq 1$ are comaximal, so there exists $x_i \in A_1$ and $y_i \in A_i$ such that $x_i + y_i = 1$. Now, $x_i + y_i \equiv y_i \pmod{A_1} \in R/A_1$. Then,

$$\begin{aligned} 1 &= (x_2 + y_2) \dots (x_k + y_k) \\ &= \text{terms containing } x_j + y_2 \dots y_k \\ &\in A_1 + A_2 \dots A_k. \end{aligned}$$

So, A and B are comaximal. □

Corollary 2.76. *Let n be a positive integer such that its prime factorization is $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$. Then,*

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/p_1^{\alpha_1}\mathbb{Z} \times \dots \times \mathbb{Z}/p_k^{\alpha_k}\mathbb{Z},$$

and

$$\mathbb{Z}/n\mathbb{Z}^\times \cong \mathbb{Z}/p_1^{\alpha_1}\mathbb{Z}^\times \times \dots \times \mathbb{Z}/p_k^{\alpha_k}\mathbb{Z}^\times.$$

Proposition 2.77 (CRT and Idempotents). *An element $x \in R$ is called an idempotent if $x^2 = x$. Suppose x is an idempotent and assume that $xr = rx$ for all $r \in R$. Then,*

1. Rx and $R(1 - x)$ are ideals of R ,
2. $R \cong Rx \times R(1 - x)$, and
3. x is the identity of Rx and $(1 - x)$ is the identity of $R(1 - x)$.

Problem 2.78. Find $a \in \mathbb{Z}$ such that $a \equiv 1 \pmod{19}$ and $a \equiv 2 \pmod{17}$.

Answer. Note that 17 and 19 are coprime, so they are comaximal. Then, we can use CRT: $\mathbb{Z}/(17 \cdot 19) \cong \mathbb{Z}/(17) \times \mathbb{Z}/(19)$. Now, $a \equiv 1 \pmod{19}$, then $a = 19k + 1$ for some integer k . On the other hand,

$$\begin{aligned} a = 19k + 1 &\equiv 2 \pmod{17} \\ 2k + 1 &\equiv 2 \pmod{17} \\ 2k &\equiv 1 \pmod{17}. \end{aligned}$$

Now, in $\mathbb{Z}/17\mathbb{Z}$, the multiplicative inverse of 2 is $\bar{9}$. Note that taking $k = 9$, we get $a = 19(9) + 1 = 172$ which is 1 modulo 19.

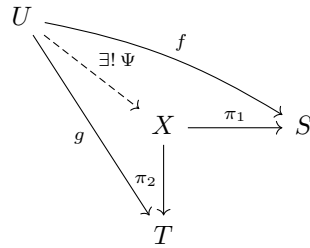
Problem 2.79. Find the smallest integer $m > 0$ such that $a^m \equiv 1 \pmod{105}$ for all a coprime to 105.

Answer. Note that $105 = 3 \times 5 \times 7$. Using CRT, we get that $(\mathbb{Z}/(105))^\times \cong (\mathbb{Z}/(3))^\times \times (\mathbb{Z}/(5))^\times \times (\mathbb{Z}/(7))^\times$. If a is coprime to 105, then $\gcd(a, 105) = 1$, so a is a unit on the left hand side, and so it maps to units in the three rings on the right hand side. In particular, $a^2 \equiv 1 \pmod{3}$, $a^4 \equiv 1 \pmod{5}$ and $a^6 \equiv 1 \pmod{7}$. Then, the smallest integer m that is a multiple of 2, 4, and 6 is 12.

2.10. PRODUCTS AND FIBRE PRODUCTS

We will, first, define the product of rings more abstractly via a universal property.

Definition 2.80. Given rings S and T , the product of the rings is a ring X equipped with morphisms $\pi_S : X \rightarrow S$ and $\pi_T : X \rightarrow T$ that satisfies the following universal property: if U is another such ring with morphisms $f : U \rightarrow S$ and $g : U \rightarrow T$, then there exists a unique morphism $\Psi : U \rightarrow X$:



Note that $X \cong S \times T$. Firstly, observe that by the universal property of X , there exists a unique morphism $\Psi_{\rightarrow X} : S \times T \rightarrow X$. By the universal property of $\Psi_{X \rightarrow} : X \rightarrow S \times T$. Now, we can show that $\Psi_{\rightarrow X} \circ \Psi_{X \rightarrow} = \text{id}_X$. In the diagram above, replace U with X , and replace Ψ with $\Psi_{\rightarrow X} \circ \Psi_{X \rightarrow}$. However, since the morphism must be unique while id_X also makes the diagram commute, it must be so that $\Psi_{\rightarrow X} \circ \Psi_{X \rightarrow} = \text{id}_X$. Similarly, we can show that $\Psi_{X \rightarrow} \circ \Psi_{\rightarrow X} = \text{id}_{S \times T}$. As such, $X \cong S \times T$.

Definition 2.81 (Fibre Products). Let R , S , and T be rings, and let $\alpha : S \rightarrow R$ and $\beta : T \rightarrow R$ be ring morphisms. Then, $S \times_R T$ is a ring, called the fibre product, equipped with two morphisms $\varphi_S : S \times_R T \rightarrow S$ and $\varphi_T : S \times_R T \rightarrow T$, such that $\alpha \circ \varphi_S = \beta \circ \varphi_T$ (as shown in the diagram below which commutes):

$$\begin{array}{ccc} S \times_R T & \xrightarrow{\varphi_T} & T \\ \varphi_S \downarrow & & \downarrow \beta \\ S & \xrightarrow{\alpha} & R \end{array}$$

The fibre product also, by definition, satisfies the following universal property: if U is another ring with $f : U \rightarrow S$ and $g : U \rightarrow T$ such that $\alpha \circ f = \beta \circ g$, then there exists a unique ring morphism $\Psi : U \rightarrow S \times_R T$:

$$\begin{array}{ccccc} U & & & & \\ & \searrow g & & & \\ & & S \times_R T & \xrightarrow{\varphi_T} & T \\ & \swarrow f & \downarrow \varphi_S & & \downarrow \beta \\ & & S & \xrightarrow{\alpha} & R \end{array}$$

$\exists! \Psi$

Note that the fibre product is defined with respect to not only the rings but also the morphisms. Therefore, a more precise notation would be $S \times_{R, \alpha, \beta} T$, since the fibre product is not independent of $\alpha : S \rightarrow R$ and $\beta : T \rightarrow R$.

Example 2.82. Consider the ring $\mathbb{Z} \times \mathbb{Z}$. Then, $\mathbb{Z} \times_{\mathbb{Z}/3\mathbb{Z}} \mathbb{Z}$ is a subring of $\mathbb{Z}/3\mathbb{Z}$. The elements in the fibre product are:

$$\mathbb{Z} \times_{\mathbb{Z}/3\mathbb{Z}} \mathbb{Z} = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a \equiv b \pmod{3}\}.$$

Proposition 2.83. Let I and J be ideals in R . Then,

$$R/(I \cap J) \cong (R/I) \times_{R/(I+J)} (R/J).$$

Note that $(R/I) \times_{R/(I+J)} (R/J) = \{(r + I, s + I) \in R/I \times S/I : r + I + J = s + I + J\}$

As a generalization of the fibre product, we have the *limit* in category theory:

Problem 2.84. Suppose that $\mathcal{R}$ is a set of rings and that $\mathcal{M}$ is a set of homomorphisms between elements of $\mathcal{R}$. We call a ring R together with a homomorphism $\phi_S : R \rightarrow S$ for all $S \in \mathcal{R}$ a *limit* (or *inverse limit*) if: (1) for each $\alpha : S \rightarrow T$ in $\mathcal{M}$, $\phi_T = \alpha \circ \phi_S$, and (2) for any other $(R', \{\phi_{S'} : R' \rightarrow S'\}_{S' \in \mathcal{R}})$ that satisfies these requirements, there exists a unique $\Psi : R' \rightarrow R$ such that $\phi_{S'} = \phi_S \circ \Psi$ for all $S \in \mathcal{R}$. Then, $(\mathcal{R}, \mathcal{M})$ has a limit.

Proof. First, observe that if S and T are rings with $\alpha : S \rightarrow T$, then the fibre product $S \times_T T$ satisfies:

$$\begin{array}{ccccc} U & & & & \\ & \searrow g & & & \\ & & S \times_T T & \xrightarrow{\varphi_T} & T \\ & \swarrow f & \downarrow \varphi_S & & \downarrow \beta \\ & & S & \xrightarrow{\alpha} & T \end{array}$$

$\exists! \Psi$

Now, we explicitly construct the fibre product. Note that since $\alpha \circ \phi_S = \text{id} \circ \phi_T = \phi_T$, we define the fibre product with projection maps such that after composing with α and id , the morphisms agree:

$$S \times_T T = \{(s, t) \in S \times T \mid \alpha(s) = t\},$$

where

$$\phi_S(s, t) = s, \phi_T(s, t) = t.$$

Now, for an arbitrary set of rings and morphisms between the rings, we can define the fibre product:

$$X := \{(x_S)_{S \in \mathcal{R}} \mid \forall \alpha_{ST} : S \rightarrow T, \alpha_{ST}(x_S) = x_T\}.$$

Now, we must verify all the important requirements.

X is a ring. Clearly, $0 = (0, \dots, 0) \in X$ since 0 is in each ring $S \in \mathcal{R}$ and every ring morphism maps 0 to 0 . Similarly, $1 = (1, \dots, 1) \in X$ since 1 is in each ring. We can define addition and multiplication as:

$$(x_S)_{S \in \mathcal{R}} + (x'_S)_{S \in \mathcal{R}} = (x_S + x'_S)_{S \in \mathcal{R}}$$

and

$$(x_S)_{S \in \mathcal{R}} \cdot (x'_S)_{S \in \mathcal{R}} = (x_S \cdot x'_S)_{S \in \mathcal{R}}.$$

In particular, the elements on the right hand side are also in X , since if $\alpha_{ST}(x_S) = x_T \in T$ and $\alpha_{ST}(x'_S) = x'_T$, then, $\alpha(x_S + x'_S) = \alpha(x_S) + \alpha(x'_S) = x_T + x'_T$. Similarly, $\alpha(x_S \cdot x'_S) = \alpha(x_S)\alpha(x'_S) = x_T \cdot x'_T$. Commutativity of addition and multiplication follows from them being respected in each ring $S \in \mathcal{R}$. Associativity and distributivity also follows from the fact that the morphisms preserve these properties: given $x_p + (x'_p + x''_p) = (x_p + x'_p) + x''_p$ in the ring P . Then, $\alpha : P \rightarrow Q$ satisfies:

$$\begin{aligned} \alpha(x_p + (x'_p + x''_p)) &= \alpha(x_p) + \alpha(x'_p + x''_p) \\ &= \alpha(x_p) + \alpha(x'_p) + \alpha(x''_p) \\ &= \alpha(x_p + x'_p) + \alpha(x''_p) \\ &= \alpha((x_p + x'_p) + x''_p). \end{aligned}$$

Distributivity is also preserved:

$$\begin{aligned} \alpha(x_p \cdot (x'_p + x''_p)) &= \alpha(x_p) \cdot \alpha(x'_p + x''_p) \\ &= \alpha(x_p) \cdot (\alpha(x'_p) + \alpha(x''_p)) \\ &= \alpha(x_p) \cdot \alpha(x'_p) + \alpha(x_p) \cdot \alpha(x''_p) \\ &= \alpha(x_p \cdot x'_p) + \alpha(x_p \cdot x''_p) \\ &= \alpha(x_p \cdot x'_p + x_p \cdot x''_p). \end{aligned}$$

Constructing each morphism $\phi_S : X \rightarrow S$. We define these morphisms to be simply the projections:

$$\phi_S((x_P)_{P \in \mathcal{R}}) = x_S.$$

Then, for any $\alpha : S \rightarrow T$ such that $\alpha(x_S) = x_T$, with $(x_P)_{P \in \mathcal{R}}$, $(\alpha \circ \phi_S)(x_P)_{P \in \mathcal{R}} = \alpha(x_S) = x_T$. On the other hand, $\phi_T((x_P)_{P \in \mathcal{R}}) = x_T$, so $\alpha \circ \phi_S = \phi_T$. Finally, suppose R' is another ring with morphisms $\phi'_P : R' \rightarrow P$ satisfying these properties, then construct the morphism

$$\Psi : R' \rightarrow X$$

by

$$\Psi(r') = ((\phi'_P(r'))_{P \in \mathcal{R}}).$$

This is a well-defined map because each ϕ'_P is well-defined and it is a ring morphism since each ϕ'_P is a ring morphism. Then,

$$\phi_S \circ \Psi(r') = \phi_S(((\phi'_P(r'))_{P \in \mathcal{R}})) = \phi'_S(r')$$

for all $r' \in R'$. So, $\phi'_S = \phi_S \circ \Psi$.

Finally, we show uniqueness: suppose $\varphi : R' \rightarrow X$ is another such morphism. Then,

$$(\phi_S \circ \varphi)(r') = \phi'_S(r')$$

for each $r' \in R'$ and each $S \in \mathcal{R}$. Since ϕ_S is a projection, $\varphi(r')$ must have $\phi'_S(r')$ in the S -th coordinate. Since this is true for each ring S in $\mathcal{R}$, therefore

$$\varphi(\cdot) = ((\phi'_S(\cdot))_{S \in \mathcal{R}}) = \Psi(\cdot).$$

□

2.11. SPECIAL INTEGRAL DOMAINS

We now at some important types of integral domains.

2.11.1 Euclidean Domains

Definition 2.85 (Norm). *Let R be an integral domain. Then, any function $N : R \rightarrow \mathbb{Z}_{\geq 0}$ with $N(0) = 0$ is called a norm on the integral domain R . If $N(x) > 0$ for all $x \neq 0$, then N is called a positive norm.*

Definition 2.86 (Euclidean Domains). *Let R be an integral domain. R is called a Euclidean Domain if there exists a norm N on R such that for all $x, y \in R$ such that $y \neq 0$, there exists $q, r \in R$ such that*

$$x = qy + r, \text{ where } r = 0 \text{ or } N(r) < N(y).$$

We say q is the quotient and r is the remainder.

Euclidean Domains allow us to use the Euclidean algorithm.

Euclidean Algorithm Let $a, b \in R$. Then, we can apply successive “divisions”:

$$a = q_0b + r_0 \tag{3}$$

$$b = q_1r_0 + r_1 \tag{4}$$

$$r_0 = q_2r_1 + r_2 \tag{5}$$

$$r_1 = q_3r_2 + r_3 \tag{6}$$

$$\vdots$$

$$r_{n-2} = q_nr_{n-1} + r_n \tag{7}$$

$$r_{n-1} = q_{n+1}r_n. \tag{8}$$

Here r_n is the last nonzero remainder. Such an r_n must exist since

$$N(b) > N(r_0) > N(r_1) > \cdots > N(r_n)$$

is a strictly decreasing sequence of positive integers whenever the remainders are nonzero, and therefore the process cannot continue indefinitely.

Proposition 2.87 (All ideals in Euclidean Domains are principal). *Every ideal in a Euclidean Domain is a principal ideal, i.e. if $I \neq 0$ is an ideal in the Euclidean Domain R , then $I = (x)$ where $x \neq 0$ and x is the element of I with the minimum norm.*

Proof. If $I = (0)$, we are done, by setting $x = 0$. Assume that $I \neq 0$. Then, consider the set $\{N(r) \mid r \in I\} \subseteq \mathbb{Z}$. By the well-ordering of $\mathbb{Z}$, this set has an element x that has the minimum norm. Now, $(x) \subseteq I$ since $x \in I$. On the other hand, consider any $r \in I$. Then, $r = xq + d$ for some q and d in R . Now, $d = r - xq \in I$ since $r \in I$ and $xq \in (x) \subseteq I$. Then, $d \in I$ with $N(d) < N(x)$ if $d \neq 0$, which is impossible since x is the element of I with minimum norm. Therefore, $d = 0$, which shows $r = xq \in (x)$, so $I \subseteq (x)$. This shows $I = (x)$. $\square$

Examples 2.88. Here are examples and non-examples of Euclidean Domains.

1. All fields are Euclidean Domains, since for all $x, y \in F$ such that $y \neq 0$, we have that $x = qy$ for $q = xy^{-1}$. We can use the trivial zero norm: $N(r) = 0$ for all r .
2. $\mathbb{Z}$ is a Euclidean Domain with $N(x) = |x|$.
3. If F is a field, then $F[x]$ is a Euclidean Domain, with the norm $N(p(x)) = \deg p(x)$. So, $\mathbb{Q}[x]$ is a Euclidean Domain.
4. $\mathbb{Z}[x]$ is not a Euclidean Domain. We have previously shown that the ideal $(2, x) \leq \mathbb{Z}[x]$ is not a principal ideal.

Definition 2.89 (Divisors and GCD). Let R be a commutative ring and let $x, y \in R$ such that $y \neq 0$.

1. x is a multiple of y if there exists $m \in R$ such that $x = my$. We say y divides x , i.e. $y \mid x$.
2. A greatest common divisor of x and y is a non-zero $d \in R$ such that (i) $d \mid x$ and $d \mid y$, and (ii) if $d' \mid x$ and $d' \mid y$, then $d' \mid d$.

Proposition 2.90. In a commutative ring R with $x \neq 0$ and $y \neq 0$ in R , if

$$(x, y) = (d)$$

for some $d \in R$, then

$$d = \gcd(x, y).$$

Proposition 2.91. Let R be an integral domain. Then, if $x, y \in R$ generate the same principal ideal, i.e. $(x) = (y)$, then $x = yu$ for some unit $u \in R^\times$. In particular, if $x = \gcd(a, b) = y$ for some $a, b \in R$, then $x = yu$ for some unit $u \in R^\times$.

Proposition 2.92 (Euclidean Domains have GCD). Let R be a Euclidean Domain. Let $x \neq 0$ and $y \neq 0$ in R . Let r_n be the last non-zero remainder of the Euclidean algorithm on x and y . Then,

$$r_n = \gcd(x, y),$$

$$(r_n) = (x, y),$$

and

$$\exists \alpha, \beta \in R, \text{ s.t. } \alpha x + \beta y = r_n.$$

2.11.2 Principal Ideal Domains (PID)

Definition 2.93 (Principal Ideal Domains). A principal ideal domain (PID) is an integral domain in which every ideal is a principal ideal.

Examples 2.94. 1. $\mathbb{Z}$ is a PID, as we have already shown before.

2. $\mathbb{Z}[x]$ is not a PID since the ideal $(2, x)$ can not be a principal ideal.

3. All Euclidean Domains are PIDs.

4. If F is a field, then $F[x]$ is a Euclidean domain, so $F[x]$ is a PID.

5. The quadratic integer ring $\mathbb{Z}[\sqrt{-5}]$ is not a PID as the ideal $(3, 1 + \sqrt{-5})$ is a non-principal ideal.

Observation. It is possible for two non-principal ideals' product to be a principal ideal. For example, in $\mathbb{Z}[\sqrt{-5}]$, the ideals $(3, 1 + \sqrt{-5})(3, 1 - \sqrt{-5})$ is principal even though neither of the two individual ideals is principal.

Observation. All Euclidean Domains are PIDs. All PIDs are not Euclidean Domains. For example, the ring $\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$ is a PID but not a Euclidean Domain.

Proposition 2.95 (Immediate Properties of PIDs). Let R be a PID. Let $a \neq 0$ and $b \neq 0$ in R . Let $(d) = (a, b)$ as ideals. Then, $d = \gcd(a, b)$ and there exists $x, y \in R$ such that $d = ax + by$. Furthermore, d is unique up to a unit of R , i.e. if $(d) = (a, b) = (d')$, then $d = ud'$ for some unit $u \in R^\times$.

Proposition 2.96. Every non-zero prime ideal in a PID is a maximal ideal.

Proof. Suppose (p) is a prime ideal in a PID and $(p) \neq 0$. Suppose, for contradiction, $(p) \subset (q) \subset R$. Then, since $q \mid p$, i.e. there exists $s \in R$ such that $p = qs$. Then, either $q \in (p)$ or $s \in (p)$.

Case 1: $q \in (p)$. Then, $(q) \subset (p)$, so $(p) = (q)$, which is a contradiction.

Case 2: $s \in (p)$, so $(s) \subset (p)$. Consider any $r \in (p)$, i.e. there exists $\tilde{r} \in R$ such that $r = \tilde{r}p = \tilde{r}qs \in (s)$, so $(p) \subset (s)$, so $(s) = (p)$. Then, $s = pu$ for some unit u , which suggests that $p = qs = qpu$, so $p(qu - 1) = 0$. Since $p \neq 0$, and an integral domain cannot have zero divisors, therefore, $qu = 1$, so $(q) = R$, which is also a contradiction. As such (p) is a maximal ideal. $\square$

Proposition 2.97. If R is a commutative ring such that $R[x]$ is a PID or a Euclidean Domain, then R is a field.

Proof. Let $R[x]$ be a PID or Euclidean Domain. Then, $R[x]$ is an integral domain, and so R is an integral domain. As such $(x - 1)$ is a prime ideal in $R[x]$ as one can easily verify. Since $R[x]$ is a PID (or Euclidean domain), then $(x - 1)$ is a maximal ideal. Therefore, $R[x]/(x - 1)$ is a field. However, $R[x]/(x - 1) \cong R$, so R is a field.

Note that this last isomorphism follows by explicitly constructing the map: $\varphi : R[x] \rightarrow R$ by $\varphi(x) = 1$. Then, this map is clearly surjective and its kernel is $x - 1$, so $R[x]/(x - 1) \cong R$. $\square$

2.11.3 Prime Elements and Irreducible Elements

Definition 2.98 (Irreducible elements, prime elements, and associates). *Let R be an integral domain.*

1. Suppose $r \neq 0$ and r is not a unit. Then, r is called an *irreducible element* if $r = xy$ implies x is a unit and/or y is a unit. Otherwise, r is called a *reducible*.
2. $p \neq 0$ is called a *prime* if (p) is a prime ideal. Equivalently, $p \neq 0$ is called a *prime* if $p \mid xy$ implies $p \mid x$ and/or $p \mid y$.
3. The elements x and y are called *associate* if there exists a unit u such that $x = uy$.

Proposition 2.99 (In IDs, prime $\implies$ irreducible). *In an integral domain, a prime element is always irreducible.*

Proof. Suppose p is a prime element in an integral domain. Now, suppose $p = xy$ for some $x, y \in R$. Then, $xy \in (p)$, so either $x \in (p)$ or $y \in (p)$. WLOG, suppose the former. Then, $x = px'$ for some $x' \in R$. Then, $p = xy = px'y = x'yp$. Since p is not 0 and there is no zero divisor in the integral domain, $x'y = 1$, so y is a unit. $\square$

Observation. *In an integral domain, primes are irreducible, but irreducibles are not necessarily prime. An example of an irreducible that is not a prime is 3 in $\mathbb{Z}[\sqrt{-5}]$.*

Proposition 2.100. *In a PID, $x \neq 0$ is a prime if and only if it is irreducible.*

Proof. We already know that in integral domains, primes are irreducible. To show the converse, suppose x is irreducible. Now, we show that (x) is a prime ideal. Suppose, we can write $(x) \subset (m) \subset R$ where each inclusion is proper. Then, $x = mx'$ for some x' . Since x is irreducible, either m is a unit or x' is a unit. If m is a unit, then $(m) = R$, which is not possible. So x' is a unit. Then, $m = x \cdot x'^{-1}$, so $m \in (x)$, and $(m) = (x)$, which is also not possible. Therefore, (x) is a maximal ideal. We know that in any commutative ring, maximal ideals are prime ideals, so x is prime. $\square$

Example 2.101. In $\mathbb{Z}$, prime elements are irreducible.

2.11.4 Unique Factorization Domains

Definition 2.102. *A Unique Factorization Domain (UFD) is an integral domain R such that, for any non-zero $r \in R$ that is not a unit satisfies the following conditions:*

1. r can be decomposed as a product of finitely many irreducible elements: $r = r_1 r_2 \cdots r_n$, where the r_i are not necessarily distinct.
2. This decomposition is unique, i.e. if there is another decomposition $r = q_1 q_2 \cdots q_m$, then $m = n$, and after some renumbering, $r_i = q_i$.

Examples 2.103. 1. Fields are trivially a UFD (since there is no non-zero element that is not a unit).

2. Every PID is a UFD, as we will prove next.

3. If R is a UFD, then $R[x]$ is a UFD, as we will see later.

4. $\mathbb{Z}$ is a UFD and $\mathbb{Z}[x]$ is a UFD.

5. $\mathbb{Z}[2i]$ is an integral domain but not a UFD. $\mathbb{Z}[\sqrt{-5}]$ is an integral domain but not a UFD.

Proposition 2.104. *In a UFD $x \neq 0$ is a prime if and only if it is irreducible.*

Proof. We already know that in integral domains, primes are irreducibles, so we show the converse. Suppose $x \neq 0$ is irreducible. Now, suppose $ab \in (x)$. We can write $ab = xy$ for some $y \in R$. Now, after decomposition of a and b into irreducibles, suppose

$$a_1 \cdots a_n \cdot b_1 \cdots b_m = x \cdot y_1 \cdots y_k.$$

Given the decomposition of ab must be unique, that means x is associate to some a_i or b_i . WLOG, suppose x is associate to a_1 . Then, $a_1 = ux$ for some unit u . Then, $a = a_1 \cdots a_n = uxa_2 \cdots a_n \in (x)$, so x is prime. $\square$

Proposition 2.105 (GCD's exist in UFDs). *Let $a \neq 0$ and $b \neq 0$ in a UFD R . Suppose their prime factorizations are*

$$a = up_1^{e_1} \cdots p_n^{e_n}$$

and

$$b = vp_1^{f_1} \cdots p_n^{f_n}$$

where u and v are units, the primes p_i are distinct, and $e_i, f_i \geq 0$. Then,

$$d := p_1^{\min e_1, f_1} \cdots p_n^{\min e_n, f_n} = \gcd(a, b).$$

We will need the following lemma for the main result in this section:

Lemma 2.106. *In a PID, any ascending chain of ideals*

$$I_1 \subseteq I_2 \subseteq \cdots \subseteq R$$

must eventually become stationary, i.e. there exists n such that for all $k \geq n$, $I_k = I_n$.

Proof. Consider the union, $I = \bigcup_{i=1}^{\infty} I_i$. Then, we can easily show that I is itself an ideal. Given this is a PID, suppose $(x) = I$. Then, $x \in I_n$ for some n . Since this is a PID, $I_n = (a)$, and as $x \in (a) = I_n$, so $I \subseteq I_n$ which shows $I = I_n$, and so the chain becomes stationary at n . $\square$

Now, we prove the main result:

Theorem 2.107. *Every principal ideal domain is a unique factorization domain. Therefore, every Euclidean domain is also a unique factorization domain.*

Proof. Let R be a PID and let $r \in R$ be a non-zero, non-unit element. Now, if r is an irreducible, we are done. If r is not irreducible, then we can write $r = r_1 \cdot r_2$ where neither r_1 nor r_2 is a unit. Now, if both r_1 and r_2 are irreducible, we have found the irreducible decomposition and we are done. Otherwise, suppose r_1 is not irreducible. Now, we can write $r_1 = r_{11}r_{12}$. Continuing this way, we see that

$$(r) \subset (r_1) \subset (r_{11}) \subset \cdots \subset R,$$

where each ideal is generated by a reducible. Note that each inclusion must be proper. Now, this process must terminate by the previous lemma. Then, renaming the generators of each ideal, we get the decomposition $r = r_1 r_2 \cdots r_n$.

Now, we must show that this decomposition is unique. If $n = 0$, then r is, by definition, a unit. Now, if we have another decomposition $r = qs$, then q divides a unit, so it must be a unit itself, which is not possible. Therefore, the decomposition with $n = 0$ is unique. Now, suppose that $n \geq 1$. Suppose, we have another decomposition:

$$r = r_1 \cdots r_n = s_1 \cdots s_m$$

where $m \geq n$ and r_i and s_i are (not necessarily distinct) irreducible. Now, r_1 divides $s_1 \cdots s_m$, and since primes and irreducibles are equivalent in PIDs, therefore, r_1 must divide some s_i . WLOG, suppose r_1 divides s_1 , i.e. $s_1 = ur_1$. Since s_1 is irreducible, u must be a unit. Now, we cancel r_1 from both sides. We continue this way, and show that $m = n$ and each r_i is associate to s_i . $\square$

Corollary 2.108 (Fundamental Theorem of Arithmetic). *$\mathbb{Z}$ is a unique factorization domain.*

2.12. NOETHERIAN RINGS

Proposition 2.109. *Let R be a ring. The following are equivalent:*

1. *Every ideal of R is finitely generated.*
2. *Any non-empty set X of ideals in R has a maximal element (i.e. there exists some $I \in X$ such that there is no $J \in X$ such that $I \subsetneq J$).*

Proof. (1) $\implies$ (2). Let X be a collection of ideals in R . Suppose, for contradiction, that X does not have a maximal element. Let $I_1 \in X$. Then, I_1 is not maximal by our assumption, so there exists some I_2 with $I_1 \subsetneq I_2$. Continuing this way, we get a chain of ideals:

$$I_1 \subsetneq I_2 \subsetneq I_3 \subsetneq \cdots$$

Consider $I = \cup_i I_i$. Since this is the union of ideals in a chain, I is an ideal itself. Since I is finitely generated, $I = (a_1, \dots, a_m)$. Then, each $a_j \in I$ implies $a_j \in I_{n_j}$ for some $n_j \in \mathbb{N}$. Let $N = \max\{n_1, \dots, n_m\}$. Then, $a_1, \dots, a_m \in I_N$. We have that $I_N = I$, and $I_{N+1} = I_{N+2} = \cdots = I$. But $I_N \subsetneq I_{N+1}$, which is a contradiction. So, there must be a maximal element.

(2) $\implies$ (1). Consider I fixed. Define X to be the set of ideals $J \trianglelefteq R$ such that $J \subset I$ and J is finitely generated. Then, X has a maximal element J . We claim $J = I$. Suppose not; then there exists some $a \in I - J$. Then, the ideal $J + (a)$ such that $J \subsetneq J + (a) \subsetneq R$. This contradicts that J is maximal. $\square$

Definition 2.110 (Noetherian Rings). *A ring R is called Noetherian if every ideal of R is finitely generated.*

Example 2.111. Every PID is a Noetherian ring. For e.g., $\mathbb{Z}$ is Noetherian.

Example 2.112. $\mathbb{C}$ is Noetherian.

Example 2.113. $\mathbb{C}[x_1, x_2, \dots]$ is *not* a Noetherian ring. The ideal $I = (x_1, x_2, \dots)$ is not finitely generated.

Proposition 2.114 (Hilbert Basis Theorem). *If R is a Noetherian ring, then $R[x]$ is a Noetherian ring.*

Proof. Suppose R is Noetherian. Now, let $I \trianglelefteq R[x]$. We must find a finite set of elements that generate I . First, consider the set

$$J := \{\text{leading coefficients of polynomials in } I\} \cup \{0\}.$$

Since I is an ideal, J is also an ideal in R . But R is Noetherian, so there exists a finite set of elements that generate J . But each element corresponds to the leading coefficient of some polynomial in I , so, there exists a finite set of polynomials $f_1(x), f_2(x), \dots, f_r(x) \in I$ such that their leading coefficients generate J . Furthermore, consider $N \in \mathbb{Z}$ such that $N > \deg f_i$ for all $i \in \{1, \dots, r\}$.

Now, for each $m < N$, define

$$J_m := (\{\text{leading coefficients of polynomials of degree } \leq m \text{ in } I\}) \cup \{0\}.$$

Again, since J_m is an ideal in R , it must be finitely generated, so there exists a finite set of polynomials $f_{m1}, \dots, f_{ms_m} \in I$ whose degrees are $\leq m$ leading coefficients generate J_m .

Now, we claim that $I = I' = (f_{mi})_{m \in \{1, \dots, N\}, i \in \{1, \dots, s_m\}} + (f_1, \dots, f_r)$. Suppose not. Then, there exists some $g \in I$ that is not in I' . Let g be the minimal such element.

Case 1: $\deg g > N$. Now, there exists some polynomials Q_i such that $\sum_i Q_i f_i$ which has the same leading term as g , so $\deg(g - \sum_i Q_i f_i) < \deg(g)$. Since g is the minimal element that is in I but not in I' , then, $g - \sum_i Q_i f_i$ is in I' . But then, $g \in I'$.

Case 2: $n := \deg g \leq N$. Again, choose Q_j such that $\sum_j Q_j f_{nj}$ has the same leading term as g . Similar argument to case 1 shows that $g \in I'$. $\square$

2.13. POLYNOMIAL RINGS

Definition 2.115. If R is a ring and $\{x_i\}_{i \in I}$ is a collection of variables, then:

1. A monomial is $a \prod_{i \in I} x_i^{n_i}$ where $a \in R$, $n_i \in \mathbb{Z}_{\geq 0}$, and n_i is 0 for all but finitely many $i \in I$.
2. A polynomial is a sum of monomials.
3. A monic polynomial is a term whose leading term's coefficient is 1.
4. $R[x_i]_{i \in I}$ is called the polynomial ring where the operations, addition and multiplication, are defined naturally.

Definition 2.116 (Power Series). A power series, $R[[x_i]]_{i \in I}$ is the ring consisting of infinite sums of monomials. In particular,

$$R[[x]] = \left\{ \sum_{n=0}^{\infty} a_n x^n \mid a_n \in R, n \in \mathbb{Z}_{\geq 0} \right\}.$$

Definition 2.117 (Degree of Polynomials). The polynomial $p(x) = a_0 + a_1x + \cdots + a_dx^d$ is said to have degree $\deg p(x) = d$ if $a_d \neq 0$. We say that $\deg 0 = -\infty$.

Proposition 2.118. Let R be an integral domain. Then,

1. $\deg p(x)q(x) = \deg p(x) + \deg q(x)$ if $p(x) \neq 0$ and $q(x) \neq 0$.
2. $R[x]^\times = R^\times$
3. $R[x]$ is also an integral domain.
4. $R[[x]]$ is also an integral domain.

Proposition 2.119 (Universal property of polynomial rings). If $\phi : R \rightarrow S$ is a ring morphism and I is a set with the map $f : I \rightarrow S$ defined, then there exists a unique ring morphism $\Psi : R[x_i]_{i \in I} \rightarrow S$ such that $\Psi|_R = \phi$ and $\Psi(x_i) = f(i)$.

Proof. One can show this by defining $\Psi(a \prod_{i \in I} x_i^{n_i}) = \phi(a) \prod_{i \in I} f(i)^{n_i}$. Note that any polynomial can be defined using some map $\alpha : I \rightarrow \mathbb{N}$:

$$x_\alpha = \prod_{i \in I} x_i^{\alpha(i)}.$$

Then, it is easy to show that $\Psi(\sum_\alpha r_\alpha x_\alpha)$ is well-defined (here, $r_\alpha \in R$), is a ring morphism, and is unique. $\square$

The following is an extremely important proposition that we will repeatedly use in the future.

Proposition 2.120. Let R be a commutative, unital ring. Let $f(x), g(x) \in R[x]$ and suppose g is a monic polynomial. Then, there exists unique polynomials $q(x), r(x) \in R[x]$ such that $f(x) = q(x)g(x) + r(x)$, where $\deg r(x) < \deg g(x)$.

Proof. If $\deg f(x) < \deg g(x)$, then select $q(x) = 0$ and $r(x) = f(x)$ and we are done. Now, suppose $\deg f(x) \geq \deg g(x)$. Let $f(x) = a_0 + a_1x + \cdots + a_nx^n$ and let $g(x) = \cdots + x^m$ since $g(x)$ is monic. Then,

$$f(x) = a_nx^{n-m}g(x) + f_1(x)$$

for some $f_1(x)$. Since $a_nx^{n-m}g(x)$ equals the leading term of f , $\deg f_1 < \deg f$. Now, continuing inductively, we can write $f_1(x) = q_1(x)g(x) + r(x)$ where $\deg r(x) < \deg g(x)$. So, $f(x) = a_nx^{n-m}g(x) + g(x)q_1(x) + r(x)$, which tells us

$$f(x) = g(x)(a_nx^{n-m} + q_1(x)) + r(x),$$

where $\deg r(x) < \deg g(x)$. This showed existence. To show uniqueness, suppose

$$f = gq_1 + r$$

and

$$f = gq_2 + r.$$

Then, $(q_1 - q_2)g = r_2 - r_1$. If $q_1 \neq q_2$, since g is monic, $\deg g(q_1 - q_2) = \deg g + \deg(q_1 - q_2) \geq \deg g$. But $\deg r_1 - r_2 < \deg g$. This gives us a contradiction. So, $q_1 = q_2$, but then $r_2 = r_1$, and we are done. $\square$

Proposition 2.121. *If $I \trianglelefteq R$, then $R[x]/I[x] \cong (R/I)[x]$. In particular, if $I \trianglelefteq_{\text{prime}} R$, then $I[x] \trianglelefteq_{\text{prime}} R[x]$.*

Proof. The first part can be proven using the morphism: $\varphi : R[x] \rightarrow (R/I)[x]$ which sends $\varphi(a) = a + I$ and $\varphi(a_i x^i) = (a_i + I)x^i$. Then, its kernel is $I[x]$ $\square$

Example 2.122. Let $R = \mathbb{Z}$ and consider the ideal $n\mathbb{Z} \trianglelefteq \mathbb{Z}$. Then, $\mathbb{Z}[x]/n\mathbb{Z}[x] \cong \mathbb{Z}/n\mathbb{Z}[x]$. In particular, if p is prime, then $\mathbb{Z}/p\mathbb{Z}$ is a field and so $\mathbb{Z}/p\mathbb{Z}[x]$ is an integral domain.

Observation. *If $I \trianglelefteq_{\text{maximal}} R$, then $I[x]$ need not be a maximal ideal in $R[x]$. However, we do have the following guarantee:*

Proposition 2.123. *If $I \trianglelefteq_{\text{maximal}} R$, then $(I, x) \trianglelefteq_{\text{maximal}} R[x]$.*

Proof. Suppose $(I, x) \subsetneq J \subsetneq R[x]$. Now, suppose $p(x) \in J$ such that $p(x) \notin (I, x)$.

Case 1: $p(x) = p$ is constant. Since $p \notin I$, then, $I \subsetneq (I, p)$ in R . But I is maximal, so $(I, p) = R$. As such, $1 \in (I, p)$. Since $(I, p) \subseteq J$, $J = R[x]$. As such, all constant terms in J must be contained in I .

Case 2: $p(x) = f(x) + c$ where $f(x)$ consists of only non-constant terms. Now, $f(x) \in (x)$. Then, $f(x) + c + (-f(x)) = c \in J$ since $-f(x) \in (x) \in J$. But since c must be in I from case 1, $p(x) = f(x) + c \in (x) + I \subset (I, x)$. $\square$

Proposition 2.124. *Given R is a commutative, unital ring,*

$$R[x]/(f(x), g(x)) \cong (R[x]/(f(x)))/(\overline{g(x)})$$

where $\overline{g(x)} = (g(x) + (f(x)))/(f(x)) \in R[x]/(f(x))$. More generally,

$$R/(a, b) \cong (R/(a))/(\bar{b}).$$

Proof. This comes from the third isomorphism theorem. $\square$

Definition 2.125 (Primitive Polynomial). *A polynomial in which the gcd of its coefficients is 1 is called a primitive polynomial.*

Proposition 2.126. *If R is an integral domain, then $R[x]$ contains infinitely many irreducibles up to associates.*

Proof. Suppose R is infinite. Then, any degree 1 polynomial $(x - a)$ for $a \in R$ is irreducible and also pairwise non-associate (there cannot be any unit u such that $x - a = u(x - b)$, since comparing coefficients gives $u = 1$). This is because $(x - a)$ is monic, so it cannot be decomposed into two factors of degree > 0 , and since f is monic, any constant factor must be 1. So we get infinitely many irreducibles. Suppose R is finite. A finite integral domain is a field, so we are looking for irreducibles in $F[x]$. Now, suppose $F[x]$ has finitely many irreducibles, $f_1(x), \dots, f_n(x)$. Note that each $f_i(x)$ must have degree > 0 , since irreducibles are not units. Now, $s(x) = f_1(x)f_2(x) \cdots f_n(x) + 1$ is a new polynomial of degree > 0 . If $s(x)$ is irreducible, we have found a new irreducible polynomial. If not, $s(x)$ be decomposed into irreducibles since $F[x]$ is a UFD. Let $g(x)$ be an irreducible factor. But $g(x)$ cannot divide any $f_i(x)$, $\forall i \in [n]$ since then $f_i(x)$ would divide 1 and become a unit. Therefore, $g(x)$ is a new irreducible polynomial. $\square$

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A. EUCLIDEAN ALGORITHM, BÉZOUT'S IDENTITY

Euclidean Algorithm The Euclidean Algorithm is an algorithm for finding the greatest common divisor of two non-zero integers a and b .

$$a = q_0b + r_0 \tag{9}$$

$$b = q_1r_0 + r_1 \tag{10}$$

$$r_0 = q_2r_1 + r_2 \tag{11}$$

$$r_1 = q_3r_2 + r_3 \tag{12}$$

$$\vdots$$

$$r_{n-2} = q_nr_{n-1} + r_n \tag{13}$$

$$r_{n-1} = q_{n+1}r_n. \tag{14}$$

Here r_n is the last nonzero remainder. Such an r_n must exist since

$$|b| > |r_0| > |r_1| > \cdots > |r_n|$$

is a strictly decreasing sequence of positive integers whenever the remainders are nonzero, and therefore the process cannot continue indefinitely. The final nonzero remainder r_n is the greatest common divisor of a and b , i.e.,

$$\gcd(a, b) = r_n =: (a, b).$$

Bézout's Identity Let $a, b \in \mathbb{Z}$, not both zero. Then there exist integers x and y such that

$$ax + by = \gcd(a, b).$$

In other words, the greatest common divisor of a and b can be written as an integer linear combination of a and b .

Proof. Suppose $d = \gcd(a, b)$. Then, $a = pd$ and $b = qd$ for some $p, q \in \mathbb{Z}$. Now, p and q must be relatively prime since, otherwise, their greatest common divisor could be absorbed into d . So, all we need to do is prove the statement for when a and b are relatively prime now.

Suppose a and b are relatively prime. Consider the set $S = \{ax + by \mid x, y \in \mathbb{Z}\}$. By the well-ordering principle, S has a smallest element. Suppose the smallest element is $m = ax_0 + by_0$. Then, we must show that $m \mid a$ and $m \mid b$. Using the division algorithm on a with divisor m , we get that $a = qm + r$ where $0 \leq r < m$. However, $r = a - qm = a - q(ax_0 + by_0) = a(1 - qx_0) + b(-qy_0) \in S$ and is smaller than m , so r must be 0. Similarly, m divides b . Therefore, m must be 1. $\square$